Learning Distributed Coloring Rules: Functional Success, Continuations, and Transfer

Research collaboration

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Abstract

What does successful execution reveal about a learned distributed algorithm? We study local neural decision modules inside explicit coloring interpreters. Seven scoped teachers and three architectures yield 21 functionally passing combinations, each with five checkpoints and 1,000 successful runs per checkpoint under heterogeneous protocols. Frozen transfer/control conditions give 449,791 successful runs out of 450,000 in the primary cohort, with failures in two checkpoint–condition cells. Against this positive functional context, we define typed continuation contracts and show how perfect measured native agreement can leave responses to alternative messages undetermined. Separate paired studies expose the role of acquired support: counterfactual supervision improves some continuation measures while sometimes reducing autonomous completion. We distinguish occurrence, input-type and execution-relevant errors rather than treating one aggregate accuracy as algorithm identification. Formal transition statements and component interventions rescue 1,799 selected CF failures by replacing COMMIT alone and a different 796 failures by replacing participation alone; transfer replays distinguish unsafe acceptances from correct vetoes that leave progress or scheduled repair unresolved. A neighbor-message follow-up provides a separate positive example of correcting a context-coverage defect. The contribution is a scoped evaluation of learned local rules and their failure mechanisms, not a universal correctness or training-superiority claim.

1 Introduction

A neural policy can produce valid colorings without reproducing every decision rule of its teacher. Conversely, a policy can reject a bad proposal correctly and still fail to finish. We study these distinctions in a setting where learned local modules are combined with declared communication, decoding and scheduling interfaces. The policies work broadly on finite benchmarks: all seven teacher variants have passing packages for each of three architectures, and sixteen of eighteen extensible combinations pass all five selected transfer conditions in all five training seeds.

Functional success is therefore the starting point for asking what was learned. In one comparison, three BE/E1 procedures match every measured native action but respond differently when a participation message or queried option changes. The off-native reference is a specified continuation contract, not an intention uniquely recoverable from a successful trace. In another comparison, equal-type continuation accuracy improves in all ten BE seed pairs while autonomous success decreases. The errors have different consequences: a confirmation at a silent root cannot write, whereas rejecting a mandatory KW recoloring violates the next bound. Under frozen transfer, two adjacent vertices can repeatedly propose the same color and correctly veto one another; a speculative repair can likewise be correctly vetoed without resolving the existing conflict.

Our contribution is an empirical study of these relations with three connected parts. First, we give scoped teacher interfaces, explicit learned and supplied modules, native functional coverage and frozen transfer. Second, we define continuation contracts and compare supervised procedures with class, support and type accounting, retaining uncertain and adverse outcomes. Third, we connect continuation errors to execution through precise transition statements, conditional progress arguments and replayable component interventions. The resulting question is which acquired local responses matter on reached executions, rather than whether one global accuracy identifies an algorithm. Elementary support arguments clarify the question; they are not claimed as new general learning lower bounds.

Relation to previous work. Neural graph coloring predates this study (Lemos et al., 2019), as do graph-algorithm execution with MAX aggregation (Veličković et al., 2020) and trace-based reasoning benchmarks (Veličković et al., 2022) within neural algorithmic reasoning (Veličković and Blundell, 2021). DAgger addresses learner-induced state distributions (Ross et al., 2011); causal confusion in imitation already motivates interventions (de Haan et al., 2019). Our fixed-bank teacher continuations are not a new formulation of these general ideas. GNN/local-computation connections and information restrictions are established (Loukas, 2020; Sato et al., 2019). SALSA-CLRS already includes distributed and randomized algorithms (Minder et al., 2023). Alignment and schedules affect extrapolation (Xu et al., 2020, 2021; Mathys et al., 2023); asynchronous robustness requires its own assumptions (Solodova et al., 2025), which we do not test. Recent coloring methods learn parallel orderings (Langedal and Manne, 2025), use recursive warm starts (Vanderbush and Weber, 2026), or learn contrastive coloring geometry with transfer (Le et al., 2026). Recent sufficient conditions for provable algorithm learning (Wittig et al., 2026) go beyond our finite extrapolation evidence. Exact local-instruction learning under bounded-degree and finite-precision assumptions is studied by Qharabagh et al. (2026); discrete neural reasoning already uses finite transition verification (Rodionov and Prokhorenkova, 2025). Our conditional MAX certificate and finite transfer tests do not establish those general guarantees. Hint-ReLIC already regularizes algorithmic predictions across input augmentations preserving the next trace step (Bevilacqua et al., 2023); we instead acquire explicit teacher-continuation labels under alternative messages, whose targets can change. Interchange-intervention training and causal distillation already match high-level counterfactual computations (Geiger et al., 2022; Wu et al., 2022); our alignment is fixed at a typed message interface and is not a new causal-training paradigm. No-hint reasoning already separates terminal learning from trajectory imitation (Rodionov and Prokhorenkova, 2023). Mechanistic gaps between effective and faithful execution are studied in Saldyt and Kambhampati (2025); concurrent work finds failures despite fitting every training hint (Najafgholizadeh and Afsharchi, 2026). Our specific increment is the joint accounting of contract, acquired support, reached error strata and safety/progress mechanisms for these distributed coloring modules. Appendix K compares candidate contributions with precedents.

2 Interfaces and Continuation Targets

We use finite simple undirected labeled graphs $G = (V, E)$, $n = |V|$, $d(v) = |N(v)|$ and maximum degree Δ. Native training uses $\Delta \le 3$ and $C_v = \{0, 1, 2, 3\}$. Transfer uses either the global list $C_v = \{0, \dots, \Delta\}$ or local initial segments $C_v = \{0, \dots, d(v)\}$. These are not arbitrary list-coloring instances. Partial colors satisfy $x_v \in \{-1\} \cup C_v$, where -1 is pending. Availability is $\mathcal{A}_v = C_v \setminus \{x_u : u \in N(v), x_u \neq -1\}$. For a pending vertex, $U_v = N(v) \cap \{u : x_u = -1\}$ and $d_U(v) = |U_v|$.

A teacher is a specified state machine with observations, private inputs, phases and an interpreter. The learned policy receives declared observations and emits decisions; the interpreter applies them. Anonymous trials use two neighbor exchanges; ColorTrials communicates availability-dependent preferences in the second. JP and Greedy need one exchange mathematically, but their neural decomposition retains two. GM has supplied blocks, barriers and a serial repair queue; BE supplies algebraic predicates and a reduction clock. These are distinct computational regimes, not equal-cost anonymous LOCAL algorithms. Message length and palette computation grow with palette size; we make no CONGEST complexity claim. Recorded native steps count state-machine applications. BE has two point-reduction steps and thirteen KW steps; neural point decisions use the first exchange, whereas KW evaluates the second stage as well. Fifteen native steps are not a fifteen-round learned communication complexity bound.

Mathematical random statements use fresh independent ideal draws. Implementation uses namespaced PCG64 streams and float32 observations. Required IDs are fixed distinct integers in $\{0, \dots, 65535\}$. The implementation restricts $1 \le n \le 65536$, including anonymous experiments for convenience; this is not an anonymous-algorithm size lower bound.

Functional success, continuation fidelity and transfer. For an input/tape distribution D, interpreter I and horizon H, let $F_\pi(D; I, H)$ be the probability of a complete proper in-list coloring within H, satisfying the protocol's checked invariants. A cutoff is a benchmark failure, not proof of nontermination. For an admissible rooted query distribution Q, let $C_{\pi,T}(Q) = \mathbb{E}_Q \mathbf{1}[g_\pi(S, p, c) = O_T(S, p, c)]$, where g_π is the queried confirmation module. Class balancing and equal-type weighting define different Q weights and are reported

separately. Transfer asks whether functional success or continuation fidelity persists under a specified change in D or Q, with fixed learned tensors and any declared input adapter. It is an evaluation axis for either property, not a third independent output criterion. Action fidelity compares common states/private inputs. Transition fidelity compares updated states; different action codes need not yield different transitions. Continuation fidelity evaluates a declared oracle $O_T(S, a)$ after a typed message or color-query intervention. Such messages need not be sampled by the native teacher. Execution costs use teacher-specific units. Distributional fidelity concerns the full terminal vector and status on a fixed labeled graph. Terminal validity alone does not imply any of the other properties.

A continuation is a specified target, not an inferred intention. Fix a legal teacher-prefix state S, including its phase, permissions w, IDs, slots, private inputs and derived algebraic data. After the first exchange, a rooted query supplies typed messages (p_u, c_u) at the root and its neighbors: $p_u \in \{0, 1\}$, $p_u \le w_u$, and a color query $c_u \in C_u$ (four option indices for KW). The acquired continuation oracles implemented here use the four-color native domain; the transfer evaluator separately extends the six non-BE rules to the declared palette lists. Let $F_v(c)$ mean that the queried color, or the decoded receiver-relative KW option, is unoccupied. Define

$$E_v = \begin{cases} w_v\, \mathbf{1}[\text{no pending neighbor has a larger ID}], & T = \text{JP}, \\ w_v\, \mathbf{1}[a_v \ge q], & T = \text{BE in KW}, \\ w_v, & \text{otherwise.} \end{cases}$$

The binary confirmation oracle is

$$O_T(S, p, c) = E_v p_v F_v(c_v)\, R_T(S, p, c), \tag{1}$$

where R_T is the rule-specific condition below. It returns a confirmation bit; it does not select a new color, repair an invalid proposal, or promise progress. Its agreement with the native decision is checked when p, c are the native messages. Off-policy targets are defined by this extension of the rule, rather than uniquely identified by the original native trace.

Rule	R_T and separate targets	Legal state and permitted intervention
Luby, Johansson	No active neighbor offers c_v	Proper partial state; replace typed participation/candidate messages after first exchange
JP	No active neighbor offers c_v	Proper partial state; fixed coherent distinct IDs. New ID assignments precede replay, not independent edge edits
ColorTrials	Root preference for c_v strictly exceeds every writable neighbor's preference when that color is free there	Proper partial state; preferences and their availability-dependent messages stay coherent. Silencing p_u does not remove that neighbor's preference message
BE	$R_T = 1$ in KW. LINIAL has a separate legal-point target	Proper auxiliary coloring from validated reduction prefix; change participation or option query, not polynomial evaluations, equalities, clock or bounds
GM, Greedy	$R_T = 1$ in color phases. GM DETECT separately marks the smaller-ID endpoint of each equal-color/equal-slot edge	Scheduled prefix; GM allows legal temporary same-slot conflicts. Repair queue follows marks. Replace P/C without independently editing marks or queue

Table 1: Self-contained continuation contracts. The main experiments use rooted message queries, not arbitrary edited global states. Alternative prefixes must satisfy the teacher's native safety/schedule constraints.

For ColorTrials the competing preference is -1 when the neighbor is not writable or its queried color is occupied; strict comparison includes ties as rejection. For GM, F_v is computed against actual speculative

colors, including during repair. DETECT does not use a P/C continuation label. For BE, a queried index c decodes to $(B-1-a_v)4+c$ at a high auxiliary color, with $q=\lceil 4B/5\rceil$. The contract rejects low-color recoloring; this is learned, not enforced by the interpreter. LINIAL's native target is the first unblocked evaluation point; the interpreter computes the corresponding polynomial value. This supplied arithmetic is not learned. JP's equal-offer veto is part of the chosen continuation extension: adjacent native eligible maxima cannot occur, so this additional veto is redundant on native messages and is not an extra rule attributed to the original JP algorithm.

The continuation-query banks use exact teacher-prefix states. The component replays below declare an additional diagnostic domain: the same KW predicate is evaluated on currently reached proper auxiliary states with the actual reduction clock and bounds. Such states need not follow the exact teacher's earlier choices. This extension is explicit; its rollout distribution is separate from the teacher-prefix query distribution.

Worked KW continuation. A replayed training witness has $B=7$, $q=6$, root auxiliary color $a_v=6$ and neighbor auxiliary colors $3,0,1$. Its four decoded options are $0,1,2,3$, so only option index 2 is free. With $w_v=p_v=1$, $O_{\mathrm{BE}}(S,p,c_v=2)=1$ and $O_{\mathrm{BE}}(S,p,c_v=3)=0$; setting $p_v=0$ also gives zero. The state, clock and neighbor equalities are identical in the three queries. Native imitation sees the successful choice 2; it need not constrain rejection of choice 3 or silence. The source is seed 94101, root 0, immediately before native transition 13 (zero-based frame 12), tape 14094519662940050106. The stored reduction snapshots support replay from the second Linial stage through KW; reconstructing the first Linial stage additionally requires the original ID assignment.

Native support and a chosen extension. Let $\mathcal{X}_T$ be the admissible rooted COMMIT inputs for a fixed interface, $\mathcal{N}_T\subseteq\mathcal{X}_T$ its native-message support, and $\mathcal{H}$ a class of binary local rules. A finite observed bank $\widehat{\mathcal{N}}_T\subseteq\mathcal{N}_T$ is a different object.

Proposition 1 (Native-support underdetermination)**.** *If $g_0,g_1\in\mathcal{H}$ agree on $\mathcal{N}_T$ and differ at an admissible $x^\star\notin\mathcal{N}_T$, even complete native input/label observations cannot uniquely identify their extension to $\mathcal{X}_T$. For the implemented BE KW, GM color and Greedy rules, native messages satisfy $O_T=p_v$. Thus, whenever $\mathcal{H}$ contains both $g_0=p_v$ and $g_1=E_vp_vF_v(c_v)$ and a legal active occupied query exists, the stated ambiguity occurs.*

Proof. Both rules induce identical labels on every possible native input, hence identical native observation laws. Their different answers at $x^\star$ cannot be selected from those observations alone. For the named teachers, each native participating color root is eligible and chooses a free option; a nonparticipating root has target zero. Therefore $O_T=p_v$ natively. In the worked KW state, active option 3 is occupied: $g_0=1$ and $g_1=0$. A full exact prefix also reaches these values on a star with legal IDs $(6,3,0,1)$: both point reductions choose point 0 and preserve these constant-polynomial values; all KW steps before bound 7 leave them unchanged. A scheduled GM/Greedy root adjacent to an already colored vertex similarly admits the query of that occupied color. These queries are allowed by the contract but are not native proposals. □

This is an elementary support argument, not a new impossibility theorem for restricted hypothesis classes or learning algorithms. A restrictive prior can select an extension. We specify O_T as our reference contract; we do not infer a teacher's unique off-support intention. The empirical agreement results concern $\widehat{\mathcal{N}}_T$, not exhaustive native support.

3 Teachers and Learned Communication

Names identify the implemented variants. Bibliographic names identify source families, not guarantees imported without checking the variant. Appendix E specifies rules and conditional guarantees.

Shared communication. E1 NORMALIZED uses node/color MLPs, learned availability, private-input participation, a normalized ordered palette-prefix scorer and learned COMMIT after a second MAX exchange. DeepSets (Zaheer et al., 2017) replaces mainly the palette scorer by shared color features and mean palette context. LocalPaletteTransformer uses two-head attention (Vaswani et al., 2017) inside each node's

Teacher	Exact rule	Supplied information/computation
Luby	Participate with probability 1/2, propose uniformly available, veto equal active offers	Pending/fixed colors, two private draws; anonymous
Johansson variant	Always active, uniform available proposal, equal-offer veto	Anonymous observations; no imported triangle-free theorem
ColorTrials	Minimum strictly winning available color, else wait	Four private preferences in $\{0, \ldots, 3\}$ natively; two exchanges
JP	Pending strict ID maximum writes first-fit	Coherent relative ID comparisons; fixed IDs
BE_BASE_D3	Two polynomial reductions, thirteen KW reductions	Bounded IDs, fields $13, 7$, point/option equalities, bound/target and phase interpreter
Speculative GM	Four-block first-fit, same-slot marks, serial repair	IDs, slots, permissions, phases, queue built from predicted marks
Greedy	First-fit in increasing-ID permutation	One externally scheduled write permission per step

Table 2: Native teachers and distinct information regimes. ColorTrials is the repeated strict-preference subroutine; GM temporary conflicts are legal.

palette. Both retain learned graph communication. Functional palette positions are supplied; color-renaming equivariance is not assumed. Hidden/message widths are 64/32. Canonical E1/DeepSets/Transformer have 16,035/17,907/16,995 parameters. Native residual adapters give 19,396/21,268/20,356 where required.

Learned components include participation, proposal scores and confirmation, BE point/option decisions, and GM marks. Polynomial evaluation, schedule construction, decoding, list-membership masks and fixed-order argmax are supplied. There is no exact availability mask, winning flag, eligibility decision or collision veto in neural inference. GM's queue follows predicted marks; an exact teacher never repairs the student's rollout.

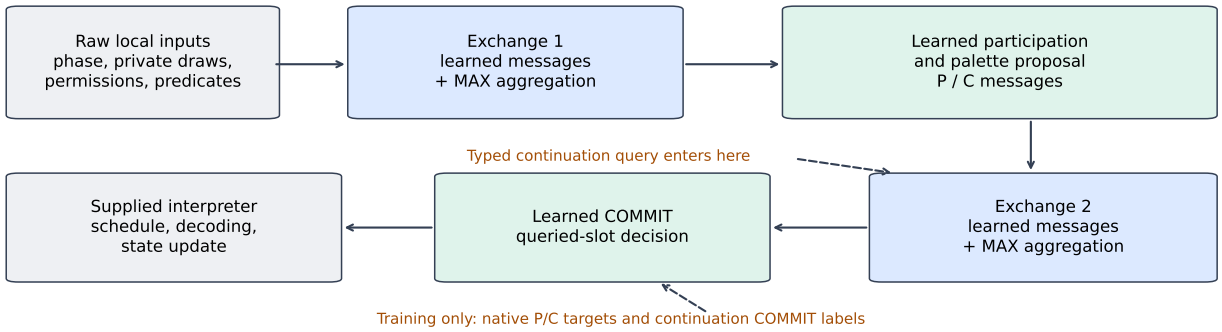


Figure 1: Two learned neighbor exchanges inside a supplied state machine. Each color slot shares parameters; graph messages use MAX. Dashed arrows locate supplied training targets and diagnostic queries. No continuation oracle or exact availability filter participates in an autonomous learned rollout.

The first exchange consumes raw pending/own-color predicates and declared phase, permission, relative-order or algebraic-equality context. Learned features feed participation and palette scores. The second exchange consumes typed P/C messages and raw color predicates, then reads the queried slot's COMMIT logit. ColorTrials instead exchanges per-color preference/features and selects its candidate after that exchange. GM DETECT queries the current color to predict a mark. Hidden-state transformations and message functions are learned; the interpreter supplies scheduling, list positions, action decoding, program color/auxiliary updates and phase/queue advancement. This decomposition locates the continuation query precisely and separates learned decisions from the externally provided computation.

Identifiers and supplied information. Coherent relative ID comparisons support priority-dependent targets; anonymous inputs cannot identify two ID assignments requiring opposite actions. Appendix D gives the elementary observation obstruction and a five-seed representation control. Failures with numerical ID magnitudes do not establish an information-theoretic inability to learn their ordering.

Variable-degree representation. The historical serializer fixes four colors and three neighbor positions. Frozen transfer evaluates the same shared tensors with a sparse structural adapter. Native parity controls cover 21,630 decisions for 150 extensible selected/control checkpoints: no hard-action differences were observed, maximum absolute logit discrepancy 1.526×10^{-5}. Separately, 1,170 native executions cover all 195 original selected/control checkpoints. These are finite checks, not bitwise-equivalence proofs for every kernel. ColorTrials uses preference/Δ in transfer, agreeing with preference/3 natively; the out-of-domain scaling choice matters. For native-context teachers, participation averages first-stage features over each receiver's valid palette slots in the structural adapter. Fixed tensors therefore still use an explicitly adapted computation outside the historical serializer; palette membership is not the only adaptation.

4 Supervision and Experimental Comparisons

Native training samples sizes 8, 12, 16, 24 at degree at most three and global $K = 4$. Historical banks use six graph families; new banks add edgeless, mixed-component and clique/isolate examples. States, observations and optimization are part of each cohort's identity.

Let P, C denote participation/candidate messages. Passive training supervises native components on teacher prefixes. Its COMMIT loss evaluates the model's current messages against a stored native label, which may be inconsistent with the correct continuation for those messages. Matched training evaluates stored teacher P, C with their concordant native label, without acquiring new labels. Counterfactual training uses the same native bank plus four stored COMMIT labels per color root: it cycles the queried root through four colors with teacher-specific message interventions and a silence case. After matched warmup, its loss is

$$L = L_P + L_C + \tfrac{1}{2} L_{\text{native,matched}} + \tfrac{1}{2} L_{\text{continuation}}.$$

LINIAL point and GM DETECT phases retain their native losses. These fixed teacher-state banks differ from collecting states visited by the learner.

Paired arms share architecture, seed, initialization, native data, minibatch order, Adam (Kingma and Ba, 2015) learning rate 10^{-3} and batch size 64; at a fixed milestone they share updates. CF adds labels and changes the confirmation loss distribution. Thus matched versus CF does not identify an intervention-only effect or efficiency at equal total label budget. Acquired component labels, model access, presentations, updates, bytes and CPU time are separate costs. Replaying a stored label is optimization, not new supervision. Shared banks are charged once for physical acquisition, separately for each model's access.

Admissibility. Anonymous queries require proper partial prefixes. JP signs derive from a coherent fixed ID order. BE states come from legal polynomial/KW replay; derived predicates cannot be edited independently. GM requires legal speculative history and schedule. A typed rooted continuation need not form a jointly consistent graph-wide intervention. We claim neither arbitrary state robustness nor joint intervention fidelity from root queries.

Historical continuation acquisition. For each color-phase root, four stored variants use option offsets $j = 0, 1, 2, 3$ modulo four relative to the native candidate. Root participation is set to permission, which can activate an ineligible BE root. In variant 3 only, an independent probability 1/4 draw silences the root. For Luby, Johansson and JP, each neighbor candidate is independently resampled from the four colors with probability 1/4 and each permitted participation bit is flipped with probability 0.15. Variant 1 then overrides those neighbor messages: every permitted neighbor participates and offers the root query. Other teachers retain native neighbor messages. All participation bits are finally masked by permission. The state, private draws/preferences, permissions, IDs, phase clock and derived equalities stay fixed; the oracle recomputes the label on the actual intervened messages. The prefilter silence probability is 1/16 across four equally counted

variants, not 1/4 overall. The separate prospective study below uses a single query per root and a different filter, and is not assigned these historical frequencies.

Selection and replication. The historical sample-budget study has seven budgets 10^3–10^6, 500 updates and five seeds: 735 policies. New procedures record 500, 1,500 and 4,000-update milestones. A prospective amendment expands development from 60 to 300 runs before new confirmation. Five candidate hashes and a final namespace are frozen before a once-only 1,000-run battery per policy. Failed candidates remain; subsequent follow-ups use new namespaces. Local registration is supported by Git/artifact provenance, not independently timestamped external preregistration. Appendix G supplies the budget/provenance table and costs.

5 Native Functional Coverage

All seven scoped teachers have passing packages for each of the three architectures: **21/21 combinations**, five selected checkpoints per combination and 1,000/1,000 proper complete invariant-valid executions per checkpoint. The 105 policies account for 105,000 successful observations. This establishes broad finite functionality within supplied interpreters; it does not establish universal correctness or equivalence of architectures.

Teacher	E1	DeepSets	Transformer
Luby	H: 5/5	H: 5/5	H: 5/5
Johansson	H: 5/5	H: 5/5	H: 5/5
ColorTrials	N: 5/5	N: 5/5	N: 5/5
JP	H: 5/5	H: 5/5	H: 5/5
BE base	N: 5/5	N: 5/5	N: 5/5
GM	N: 5/5	N: 5/5	N: 5/5
Greedy	H: 5/5	H: 5/5	H: 5/5

Table 3: Native functional coverage. Each passing checkpoint completes all 1,000 runs. H: twelve historical passive packages; N: nine later CF packages. The cohorts have different training and final-test protocols and do not constitute a homogeneous architecture-ranking experiment.

Historical packages use seeds 83101–83105; later ColorTrials/BE/GM packages use 94101–94105. Representative BE/ColorTrials policies use 4,000 updates, GM 500, while acquired-label budgets vary by package. Matched controls also pass all nine later combinations: CF is not necessary for native functional success in these tests. Final sizes are 32,48,64; seven historical versus ten later graph strata have different weights. Appendix C retains confidence-limit assumptions, unsuccessful candidates and the full budget/provenance table. The following continuation analyses concern separate historical or prospectively frozen cohorts; the 21/21 result is not attributed to every model in those analyses.

6 Native Agreement and Continuation Fidelity

We reconstruct the existing paired 4,000-update diagnostics on 60 exact teacher trajectories at sizes 28 and 36 per seed. Their graph/tape namespaces are disjoint from training. They belong to the repeatedly used development split, not an untouched final fidelity battery. BE/E1 gives 28,800/28,800 native permitted-root action agreements in every arm and seed, but very different continuation responses: The paired initial hashes, native arrays, visit counts and final minibatch RNG states agree. CF adds 3,776 labels, changes COMMIT contexts and splits its confirmation loss. This supports a difference between the full procedures, not an intervention-only effect, sample efficiency, or universal functional improvement. Only 2.7774–3.1200% of queries are positive. Every arm has zero false negatives; CF has 34 false positives, all on eligible but silent roots. Most of the raw gap concerns rejection of ineligible KW continuations. Among eligible active queries (4,150–4,598 occurrences per seed), CF is 100% correct in all seeds, whereas passive and matched accept every option, including occupied ones: mean accuracy 67.8117%, BA 50%. Thus the effect also includes learning

Procedure	Labels	Mean acc. (%)	Seed range (%)	Mean BA (%)
Passive	3,000	22.6404	9.1607–42.3297	60.1413
Matched	3,000	9.2686	9.0905–9.3970	53.2536
Counterfactual	6,776	99.9932	99.9780–100.0000	99.9965
Always reject	–	97.0465	96.8800–97.2226	50.0000
Eligibility only	–	98.5990	98.5146–98.6208	99.2781

Table 4: BE/E1, 4,000 updates. Each seed has 99,840 continuation queries. Means describe five paired training seeds; 499,200 queries per arm are correlated within states/trajectories, not independent training replicas. BA averages sensitivity and specificity within seed, then over seeds. Eligibility-only accepts iff the root is KW-eligible and participating; these semantic baselines are diagnostics, not trained competitors.

to reject occupied options, not only a redundant eligibility gate. Always rejecting is accurate overall but destroys progress.

Novelty of the evaluation queries. Reconstructing the original query RNG reproduces every numerator. Only 91 of 499,200 occurrences exactly match a complete serialized training query. However, BE COMMIT discards many serialized fields. Its queried-slot logit depends on phase, permission, signed KW offset, root participation, and the MAX-set of occupied/participating/offering neighbor types. Between 99.277% and 99.540% of diagnostic occurrences have one of these sufficient input types already in training. The four queried colors and intervention families are not held out. Among the 549, 459, 717, 522, 722 occurrences with unseen sufficient types, CF correctness is 549, 437, 717, 510, 722, versus matched 158, 342, 229, 182, 246. This posthoc restricted result preserves a seed-wise advantage. Its correlated small subsets and development reuse do not support broad new-intervention-distribution generalization. Appendix H defines the sufficient-input key and gives per-seed numerators. These unseen-type subsets include 52,108,90,56,60 positives and both classes in every seed. All three arms accept all positives; CF's advantage is again specificity on negative continuations. The mean unseen-subset BA is 99.1157% CF versus 66.5299% matched and 50% always-reject.

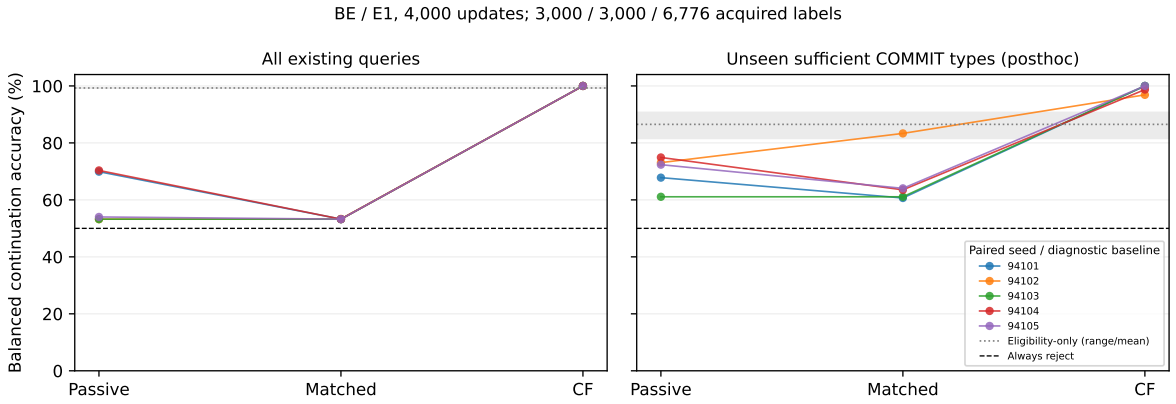


Figure 2: BE/E1 balanced accuracy across paired seeds, with constant-reject and eligibility-only diagnostic baselines. Right: posthoc unseen sufficient COMMIT types. Curves connect treatments within each paired seed; label budgets differ by arm.

Equal weight per sufficient local type. Occurrence-weighted accuracy gives frequent local configurations more weight. We additionally average accuracy within each sufficient COMMIT input type and then over distinct types, separately within each training seed. This measures the tested type set, not a natural input distribution or a new training replica. Reconstruction with the original RNG and 256-root chunks reproduces every historical numerator before this reweighting.

Procedure	Occurrence acc. (%)	BA (%)	Equal-type acc. (%)
passive	22.640	60.141	50.784
matched	9.269	53.254	46.350
counterfactual	99.993	99.996	99.488
always reject	97.046	50.000	82.317
eligibility only	98.599	99.278	78.973

Table 5: Same historical BE queries, three different estimands. Entries average five paired seeds. Equal-type accuracy weights the 115–132 distinct tested types equally within seed, rather than their 99,840 occurrences.

These estimates retain both the strong rejection difference and the limits of the familiar-type diagnostic. Per-seed seen/unseen, eligibility, silence and occupied/free strata, including all confusion counts, accompany the type-level occurrence counts in the revision evidence. The split uses the union of native and stored CF training types in all historical arms, making its novelty criterion common rather than treatment-dependent.

Other teachers and dependent architecture comparisons. At the same milestone GM/E1 has mean continuation accuracy 75.3632% for passive/matched versus 100% CF; CT has 96.6570%, 97.2439%, 99.7452%. Their denominators vary by seed. These are supporting diagnostics. Paired BE/GM CF COMMIT tensors are identical across the three palette cores at audited milestones: fixed-proposal second-stage losses do not involve their palette scorer. These are five paired confirmation replicas per milestone, not fifteen independent architecture replicas. CT couples second-stage gradients to proposal scoring and lacks that tensor identity.

The scoped answer is positive: this CF procedure better reproduces these typed teacher continuations, including some unseen local types, even when passive native agreement is already perfect. It is not always necessary, universally functionally superior, or shown more label efficient. The historical comparison remains separate from the new budget-matched experiment below.

6.1 A prospective input-context comparison

We test a separate acquisition-strategy contrast with BE/E1 and GM/E1 (19,396 parameters). Ten fresh paired seeds per teacher give forty models. Before training, a fixed SHA256 partition holds out 20% of sufficient queried-slot COMMIT type buckets, common across seeds. Phase, root/context features and distinct neighbor message types determine this key; MAX removes multiplicity. Neither training arm supervises a held-out continuation type. LINIAL and DETECT remain separate native targets. This is a split of input contexts within the same intervention templates. It does not withhold all coarser oracle-relevant configurations (quantified below).

Each pair shares 3,000 native component targets and 3,600 fresh auxiliary rooted states. On each auxiliary state the native arm acquires one native COMMIT target and CF one prespecified intervened COMMIT target. Both queries must pass the training-type filter to retain that state. GM auxiliary quotas are 1,800 ASSIGN and 1,800 REPAIR; BE uses KW only. Each model therefore accesses 6,600 optimization targets. Native P/C conditioning is supplied to both arms equally and charged separately: 7,200 root values plus retained neighbor values. Simulator computation, rejected states and duplicate queries are also recorded; this does not equate all teacher-computation costs. The root option cycles through native-option offsets 0,1,2,3 modulo four. Participation is the root permission, except that offset three independently silences it with probability 1/4; neighbor messages remain native. The cursor advances before filtering, including discarded roots. Test queries use the same intervention and retain held-out CF types; their phase quotas are 4,000 KW or 2,000 each ASSIGN/REPAIR. This single-query design differs from the historical four-query diagnostic. Appendix A.1 specifies the key, serialization, hash and acquisition order.

Adam, batch 64 and 1,500 updates are fixed before test, with 100 matched warmup updates and the same subsequent native/auxiliary loss weighting. Paired data, initialization, index sequences and first-stage final tensors agree. In BE, the paired learned procedures therefore differ only in the second exchange/COMMIT parameters, not in participation or proposal parameters. Their states can nevertheless diverge after a changed confirmation. All forty weights are frozen before generating 4,000 held-out queries per seed/teacher on sizes 28/36 and evaluating 1,000 autonomous runs per model on sizes 32/48/64. All exact references pass. Every

final continuation has zero intersection with its training sufficient-type set; both truth classes occur in every seed.

Teacher	Native BA	CF BA	Paired difference [95% interval]	Successes N / CF
BE	84.64	90.41	+5.77 [-13.73,+25.26]	9,204 / 7,405
GM	50.00	100.00	+50.00 [+50.00,+50.00]	10,000 / 10,000

Table 6: New final experiment. BA is a percentage; differences and intervals are percentage points, averaged over ten paired training seeds. Intervals bootstrap paired seeds (10,000 resamples), conditional on this fixed split/protocol; individual queries are not independent replications. Success totals are out of 10,000 autonomous runs per teacher/arm. No checkpoint selection uses these tests.

GM equal-type accuracy is 35.44% native and 100.00% CF; both arms average 26.01 native steps.

GM gains 50 percentage points in every paired seed without changing any autonomous outcome or mean step count. Its degenerate bootstrap interval reflects those identical observed seed differences, not certainty over all future types or training replicas. BE's mean difference is +5.77 points, but the interval spans zero $(-13.73, +25.26)$. BE equal-type accuracy increases from 51.70% to 86.94%, while the two endpoints weight errors differently. Equal-type accuracy favors CF in all ten paired seeds, with gains of 15–55 percentage points. Occurrence BA is heterogeneous: three native seeds score near 50%, two CF seeds near 52%, and the other paired differences are small. The mean is retained as the prespecified endpoint; it does not describe a uniform seed effect or establish an effect of zero. CF adds false negatives in two BE seeds and reduces autonomous successes from 9,204 to 7,405; one further seed has identical failures in both arms. Mean step counts over *all* final runs (14.53 native versus 13.43 CF) include early exits on failure. Every successful BE run takes 15 native steps in both arms; the lower overall mean is not a successful-run speedup. These are substantive negative results for a uniform-benefit claim.

Post-test diagnostic controls. The primary endpoint and all original outcomes above are retained. The following controls and support audit were added after inspecting the test; they are descriptive checks, not a revised primary analysis. The programmed rule $G_v = E_v p_v$ uses supplied eligibility and root participation and ignores query occupancy. It requires no training and is a diagnostic, not a learned label-efficiency competitor.

Teacher	Procedure	BA (%)	Equal-type acc. (%)
BE	Native	84.64	51.70
BE	CF	90.41	86.94
BE	Eligibility/participation	99.71	81.27
GM	Native	50.00	35.44
GM	CF	100.00	100.00
GM	Eligibility/participation	53.03	62.34

Table 7: Same final queries and sufficient-type key; means over ten seeds. The programmed diagnostic ignores occupancy. BA averages sensitivity and specificity; equal-type accuracy first averages within each distinct type, then across types within seed. The diagnostic is post-test.

For BE, the programmed rule scores 99.71% BA, above CF's 90.41%, whereas CF scores 86.94% equal-type accuracy versus 81.27%. A single negative type accounts for 95.05–96.30% of each 4,000-query bank. Class balancing therefore leaves the negative subtypes highly imbalanced; high occurrence-level BA alone does not identify the occupancy condition. Among eligible active BE queries, native and the programmed rule accept 995/995 free options but reject 0/228 occupied options. CF rejects 228/228 occupied options but accepts only 814/995 free options. These are pooled descriptive counts over ten banks, not independent training replications. Appendix A.2 reports the full confusion strata.

GM's native control receives 4,344–4,354 positive COMMIT labels and zero negative COMMIT labels in ASSIGN/REPAIR per seed, while CF receives 1,038–1,098 negatives and the test has 2,420–2,491 negatives per seed. The native model accepts all test queries, giving BA 50%; the contrast therefore demonstrates learning rejections absent from native color supervision. DETECT has separate positive/negative conflict

supervision and is excluded from these color-label counts. This result does not compare CF with another acquisition strategy supplying informative negative confirmations.

Support of the held-out queries. There are 17–23 distinct sufficient test types per BE seed and 82–92 per GM seed, despite 4,000 occurrences. For these color oracles $O_T = E_v p_v (1 - o_v)$, with o_v the queried-option occupancy. A coarser configuration (phase, E_v, p_v, o_v) is therefore oracle-sufficient. CF training has already seen configurations covering 99.475–100.000% of BE test occurrences and 100% of GM test occurrences. Native training covers only 2.125–2.900% in BE and 37.725–39.500% in GM. The same test thus has different logical-support novelty for the two acquisitions. Equal target counts do not imply equal information or equal support coverage. The original fine-key holdout remains disjoint: novelty is defined by contextual combinations rather than by this coarser configuration. There is no separate evaluation restricted to novel logical conditions. These tests do not establish extrapolation to new logical conditions. Per-seed support counts and logical overlap are in Appendix A.2; no split, label or model was changed.

Two distinct BE failure diagnoses (post-test). In the two deteriorated CF seeds, 105001 and 105008, COMMIT BA on the 3,600 acquired auxiliary queries is 99.510% and 100.000%, versus 52.756% and 52.094% on the held-out contexts. Both models visited all auxiliary queries. Failure to fit acquired labels alone does not explain these cases: the observed COMMIT gap concerns generalization across contexts, without identifying its optimization cause or an architectural impossibility. All 181 final false negatives are at the eligibility boundary $\kappa = \mathrm{sgn}(a_v - q) = 0$ and have no participating neighbor. They are distinct from the active-neighbor defect in the separate follow-up below. On eligible silent queries CF makes 81 false acceptances versus 42 for native; these bits alone are masked by participation.

Seed 105010 has a different defect: both arms complete only 204/1,000 runs, despite final COMMIT BA 99.451% native and 99.949% CF. Their shared first stage omits participation on 13/118 eligible test roots. A replay of the original 32-case cohort localizes the first invariant violation in one selected run: at KW step 7, vertex 31 must recolor as the bound drops from 21 to 17, but remains at 17 because participation is false. Its proposed value 12 is free; changing only root participation makes COMMIT accept it. Thus this state exposes an upstream omission, rather than an incorrect veto of the actual silent message. These are selected post-test diagnostics, without retuning, a continued rescue rollout or an estimate of causes across all failed runs (Appendix A.3).

The primary endpoint concerns held-out continuation types; autonomous success is secondary. This finite comparison concerns the specified acquisitions at equal target/context access. It supports neither extrapolation to new logical conditions nor an advantage over alternative informative-negative acquisition strategies. BE's occurrence-level endpoint must be read with its diagnostic baseline and type weighting; GM's control lacks negative color confirmations. No universal CF superiority, minimum sample complexity or all-source teacher efficiency follows. The protocol, forty checkpoints, all seed outcomes, per-component costs and freeze receipts accompany the new results (Appendix A).

6.2 Label budgets and neighbor-context coverage

A separate E1 study uses five paired seeds per teacher (95101–95105), 4,000 fixed updates and passive, matched and CF procedures at low and high acquired-label budgets for BE, GM and ColorTrials: 100 trainings, with 1,500-update snapshots as a secondary milestone. It uses fresh training, continuation and functional namespaces. The high native controls acquire additional native trajectories; low-budget CF removes stored continuation labels to keep its total fixed. The BE high CF/native totals are 6,776/6,774, GM high totals differ by at most two labels per seed, and ColorTrials uses 70,000/69,999. These small component-group rounding differences are recorded, rather than called exact budget equality. They remain full acquisition-procedure contrasts: the native controls receive no alternative source of informative occupied-option negatives.

CF improves standard-family BA in each BE/GM pair, including low-budget CF against high-budget native supervision. Yet the BE high-CF functional loss replicates the separation between continuation metrics and completion. Across all eight BE arms only 32/40 checkpoints pass all 1,000 functional runs; all adverse arms remain recorded. The new study's own criterion of at least 99% BA on unseen sufficient types fails: only two of five high-CF seeds reach it. BE's eligibility/participation diagnostic scores 99.26% occurrence BA. A holdout of occupied queries at late KW bounds tests a generator-level combination; it does not

Teacher	Continuation BA range (%) Matched-high	CF-high	Functional successes /5,000 Matched-high	CF-high
BE	53.25–53.30	93.81–100	5,000	4,298
GM	50	100	5,000	5,000
ColorTrials	99.34–99.39	100	5,000	5,000

Table 8: Separate label-budget study: standard-family continuation queries, five seeds, 4,000 updates. These are different models, support and endpoints from Table 6, not additional replicas of that study. Ranges are between seeds; shared queries are not independent replicas.

by itself establish new effective inputs, because queried BE COMMIT does not read the numerical bound. Received-neighbor message perturbations, unused in these training banks, also expose errors. Thus the stronger standard-family advantage does not contradict the input-context holdout result. Its 1,500-update CF snapshots already have high standard BA, so update count alone does not explain the cross-study difference.

A targeted follow-up with a new battery. After inspecting those results, a separately frozen follow-up perturbs neighbor messages in the high-CF BE bank, preserving roots, root queries and all labels. BE's oracle ignores those neighbor participation/offering messages when occupancy is fixed. Five new paired trainings use a fresh continuation and 1,000-run functional battery, together with the frozen controls. The perturbed arm accepts all positive active-free-neighbor queries in all five seeds and completes 5,000/5,000 runs. The original high-CF seed 95105 again rejects those positives and completes 301/1,000, while its perturbed counterpart completes 1,000/1,000. This is a positive result for that context intervention on the new battery, not a universal CF remedy. It addresses a different error from the $\kappa = 0$, silent-neighbor false negatives in seeds 105001/105008.

7 Which Continuation Errors Affect Execution?

Continuation error weights queries in an evaluation distribution. Functional error depends additionally on whether those queries are reached and how the interpreter uses the answer. In color phases, let z_v be the confirmation bit and $D_v(S, c_v)$ the supplied decoding of a proposed color or KW option. DETECT marks and LINIAL point reductions are separate interfaces.

Proposition 2 (Gated transition relevance)**.** *Fix a legal color-phase state S, permissions w, participation p and candidates c. Let y_v be the current persistent color (auxiliary color in KW). For the implemented interpreters,*

$$y'_v = \begin{cases} D_v(S, c_v), & w_v p_v z_v = 1, \\ y_v, & \textit{otherwise.} \end{cases}$$

Changing z only where $w_v p_v = 0$ leaves this transition unchanged. Changing z_v where $w_v p_v = 1$ changes the stored value precisely when $D_v(S, c_v) \neq y_v$. In a KW transition with new bound q, a root with $a_v \geq q$ that does not write retains a value outside $[0, q)$ and violates the checked bound invariant immediately.

Proof. The neural action decoder writes only when participation and COMMIT are true; the interpreter additionally enforces permission. It performs no eligibility or availability check. Each vertex updates its own stored value from the common old state. The displayed formula and the first two claims follow by substitution. KW lowers the bound to q regardless of whether a vertex wrote. An unchanged $a_v \geq q$ therefore violates the new bound, independently of neighbor writes. The benchmark checks that invariant after each transition and stops on violation. □

Thus a false acceptance at a silent root has no direct write effect, whereas a false rejection of an eligible active free KW option is immediately consequential. An ineligible active acceptance can change execution if an upstream participation error makes it reachable and decoding changes the stored value; eligibility alone is not an interpreter shield. These are fixed-state statements, not claims that every off-policy error is harmless or every local error causes failure in every protocol.

GM repair is a suffix property. Correct marks covering every speculative conflict edge, followed by a currently free write at every queued vertex, are sufficient for proper completion. For any initially conflicting edge, the last repaired endpoint avoids the other; an edge with no repaired endpoint was already proper. These conditions are not individually necessary. If a failed repair leaves an edge monochromatic and neither endpoint changes color later, that edge remains a final conflict. If another queued endpoint later writes a free color, it may resolve the edge. For example, three one-vertex blocks on a path with increasing IDs $1, 2, 3$ simultaneously assign color 0. DETECT marks vertices 1 and 2. Skipping repair 1 and repairing 2 to color 1 still gives $(0, 1, 0)$. This is a legal execution of the supplied schedule. A repair veto must therefore be interpreted with the remaining queue, rather than treated as a universal implication of failure.

7.1 Frozen-component interventions on reached BE states

We select all three BE seeds with any functional failure in the 40-model study, retain both arms and all 1,000 original cases per model, and compare five fixed modes: unchanged policy; exact KW confirmation on actual learned messages; exact KW participation with learned proposal/COMMIT recomputed; first-free KW candidates with learned participation/COMMIT recomputed; and all three exact KW components. Learned Linial decisions remain unchanged. This is a retrospective diagnostic on consumed cases, selected by known failures, not a new training-method test. Its protocol and source/weight hashes are frozen before replay outcomes. All baseline terminal vectors, statuses, steps and invariant flags must reproduce before interpreting interventions. The other learned modules retain their weights; their later outputs are recomputed after trajectories diverge. The exact KW predicate uses each reached state's valid auxiliary coloring, clock and bounds; it is the diagnostic state-domain extension specified with the contracts, not an assumption that every rollout state is an exact teacher prefix.

All six unchanged baselines reproduce the original 6,000 terminal vectors, statuses, step counts and invariant flags exactly. Across the 30,000 executions there are no technical errors and no newly failed cases under an intervention. In CF seeds 105001 and 105008, exact COMMIT alone raises successes from 100 and 101 to 1,000 each, rescuing all 900 and 899 failures. Changing only participation or candidates rescues none. All 1,312 and 1,322 missed valid writes reached by their unchanged policies occur at $\kappa = 0$; unlike the 181 teacher-prefix false negatives, some have participating neighbors (358 and 406). These are visit counts, not independent errors or new training replicas.

In seed 105010, both arms complete 204/1,000. Exact COMMIT or candidates alone leave all 796 failures unchanged. Exact participation alone rescues all 796 in each arm; the unchanged policy is silent on 983 eligible reached queries. All-exact KW modules yield 1,000/1,000 for each of the six models, while learned Linial modules remain fixed. Thus COMMIT-only rescues 1,799 of the 2,595 CF failures on this original battery, and participation-only rescues the remaining 796. Counts across arms share cases and first-stage tensors; they are not independent intervention replications.

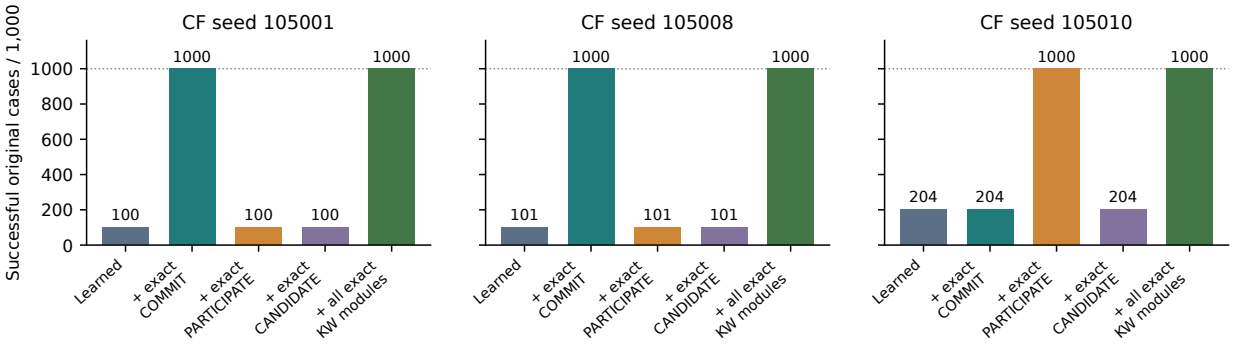


Figure 3: Component interventions on the three selected CF seeds, using all original cases. Each replacement changes one KW component and recomputes dependent learned outputs on the ensuing trajectory. All-exact replaces three KW components. These retrospective diagnostic successes do not establish a learned repair or general intervention success rate. Native arms and case-level outputs are retained in Appendix B.

The interventions use oracle decisions and are diagnostic tools, not deployable learned policies. Their fixed-battery rescue counts localize mechanisms in the selected models; they do not estimate a population remedy. Errors on teacher-prefix continuation queries and errors reached during autonomous rollouts have different denominators and are reported separately.

8 Frozen Transfer: Functional Success and Failure Mechanisms

The domain-wide totals below come from the frozen transfer campaign. Evaluation checks terminal completeness, palette compliance, edge conflicts and teacher-specific invariants. To investigate the failure mechanisms, we replay all 15 cells containing failures (15,000 executions) and apply the component interventions described below.

Teacher extensions. The six non-BE rules admit the parameterized definitions in Appendix E. JP's historical cap becomes n, preserving its priority rule. GM retains four blocks/detection/serial repair; Greedy retains its schedule. Palette lists extend the domain. CT also changes its preference alphabet to $\{0, \ldots, \Delta\}$; it is a declared parameterized variant, not identical fixed-$K4$ private inputs. BE is excluded because fixed fields and four-option blocks lack a higher-degree existence proof. Changing those constants would define another mathematical variant and interface.

Confirmatory design. After a separate pilot, weights, conditions, namespaces and horizons are frozen. All 150,000 exact-reference executions pass before students run. Ninety primary extensible checkpoints receive 1,000 executions at each of five conditions: global $n = 64, \Delta = 3, 4, 10$; local $n = 64, \Delta = 4$; global $n = 256, \Delta = 4$. Sixty native-certified matched/passive controls and thirty pre-existing CT 4,000-update controls receive the four $n = 64$ conditions. The thirty are not assumed certified. Weights stay frozen, inference uses float32 CPU. Each graph has maximum degree exactly requested. Conditions use a uniform mixture of regular, capped random, bipartite, tree, irregular, hub, disconnected, isolate-containing, heterogeneous and clique families. Capped random is not an Erdős–Rényi law. Instances/tapes are shared across policies, teachers and cores. There are 4,000 distinct labeled graphs, 20,000 tape seeds and 25,000 case–palette pairs in the 810,000 policy observations. Observation, graph and training-replica counts differ.

Raw-color reconstruction gives **808,413/810,000** correct: 1,559 collisions and 28 cutoffs in fifteen of 810 cells. This is accounting across policies, not the success rate of a mean algorithm. Primary policies give **449,791/450,000**: 209 failures in two cells, with 90,000/90,000 degree-three controls. The other 448 primary cells have zero observed failures. Secondary controls give 358,622/360,000. Pointwise checkpoint intervals describe the declared mixture, not 1,000 executions per graph family. Table 9 preserves seed ranges.

Teacher	Core	$64, 3, G$	$64, 4, G$	$64, 10, G$	$64, 4, L$	$256, 4, G$
Luby	E1	5/5	5/5	5/5	5/5	5/5
Luby	DeepSets	5/5	5/5	5/5	5/5	5/5
Luby	Transformer	5/5	5/5	5/5	5/5	5/5
Johansson	E1	5/5	5/5	5/5	5/5	5/5
Johansson	DeepSets	5/5	5/5	5/5	5/5	5/5
Johansson	Transformer	5/5	5/5	4/5 (972–1000)	5/5	5/5
ColorTrials	E1	5/5	5/5	5/5	5/5	5/5
ColorTrials	DeepSets	5/5	5/5	5/5	5/5	5/5
ColorTrials	Transformer	5/5	5/5	5/5	5/5	5/5
JP	E1	5/5	5/5	5/5	5/5	5/5
JP	DeepSets	5/5	5/5	5/5	5/5	5/5
JP	Transformer	5/5	5/5	5/5	5/5	5/5
GM	E1	5/5	5/5	5/5	5/5	5/5
GM	DeepSets	5/5	5/5	5/5	5/5	5/5
GM	Transformer	5/5	5/5	5/5	4/5 (819–1000)	5/5
Greedy	E1	5/5	5/5	5/5	5/5	5/5
Greedy	DeepSets	5/5	5/5	5/5	5/5	5/5
Greedy	Transformer	5/5	5/5	5/5	5/5	5/5

Table 9: Primary checkpoints with 1,000/1,000 successes out of five. Nonperfect cells also give the per-checkpoint success range. Columns are $(n, \Delta,$ palette$)$, G global and L local. Sixteen of eighteen combinations pass all selected conditions.

Failure mechanisms. All 15,000 replayed executions reproduce their cells' success counts, failed-case identities, failure classes and proposal/confirmation counters. Of the 1,587 failed observations across these cells, 28 Johansson cutoffs and 181 CF GM failures are progress failures without an unsafe acceptance; the other 1,378 include unsafe acceptances. This classification preserves the distinction between an invalid final state and an invalid committed proposal. The original first-failure replays additionally match complete final vectors, status and steps in their original inference batches. Johansson/Transformer 83101 has 28/1,000 degree-ten cutoffs, all on cliques. The first leaves two adjacent pending vertices with free colors $\{0, 2\}$. Both repeatedly propose color 2, correctly vetoed by COMMIT. In the preserved conditional probe each proposes 2 on all 8,192 new draws. This identifies missing observed proposal diversity in that state, not an incorrect veto or exactly zero mass for the other color.

GM/Transformer 94102 has 181/1,000 local degree-four failures in each of CF, matched and passive. In the first primary CF witness, exact assignment and detection precede a wrong repair proposal: correct rejection omits a scheduled repair and leaves a conflict. Matched/passive first witnesses include a forbidden write. Equal failure counts therefore do not imply identical local mechanisms. A separate exact-prefix control fixes $\Delta = 3$, IDs and endpoint colors but changes a degree-one endpoint's list from four to two; the Transformer misses the exact repair. This isolates palette-context sensitivity in that constructed state, not universal capacity versus optimization explanations.

One-component interventions on copied witnesses. We reconstruct each original full inference cohort from its case/tape seeds and recovered checkpoint SHA. The unchanged execution matches its stored terminal colors, steps and status, and all actions match through the selected intervention frame. At that frame only one candidate is replaced by a free alternative; all other messages and first-stage tensors stay fixed and the learned second exchange/COMMIT is recomputed. A separate control substitutes exact confirmation on the unchanged proposals once. Neither intervention changes a training run or any reported autonomous test.

These controls separate a proposal defect from an incorrect veto in the selected states. Exact-COMMIT substitution is an identity at these already correct vetoes, not an all-state oracle replacement or an audit of COMMIT everywhere. The controls do not establish that changing the proposal module would fix every failed execution, or that the first latent calibration discrepancy caused the observed terminal failure.

Eleven other failed cells involve three matched E1 CT checkpoints, including degree-three failures and incorrect simultaneous confirmations. Higher degree cannot explain all of them. Alpha-threshold discrepancies are latent calibration diagnostics, not supervised availability errors or causes by themselves. The first temporal discrepancy is not automatically sufficient to explain a final failure. Appendix I lists every cell.

A separate prospective transfer contrast. A separate fresh transfer campaign evaluates six acquisition arms (passive, matched and CF at two label budgets), GM/CT, five seeds and the 4,000-update checkpoint across four size 64 degree/palette conditions: 240,000 executions, all functionally successful. The GM policies emit no invalid proposals there, so a proposed advantage in rejecting such proposals is uninformative on this battery. CT does emit invalid proposals, but all acquisition arms contain them sufficiently to finish. This campaign uses separate models and cases and is not pooled with the historical 810,000 executions.

Exploratory scope. The separate pilot has 3,526/3,570 successes: two collisions, 42 CT resource cutoffs. It spans regular $n = 256$, degrees 2,3,4,5,10,20,50,100,200, sizes through 1,024 at degree 3 or 10, plus heterogeneous/local cases. A ten-second cap was applied only to unfinished dense CT cells after some longer degree-50 successes existed. These unequal resource horizons do not permit homogeneous success-rate ranking. One run per checkpoint/cell is exploratory. The later size pilot gives 240/240 for sixty Luby/Johansson/CT/JP checkpoints on regular $n = 2048, 4096$, degrees 3,10, one run per point. It contains four labeled graphs and twenty tapes, separately from confirmation. GM/Greedy size pilots stop at 1,024; BE stays outside this extension. Confirmation therefore reaches degree ten at size 64, and size 256 at degree four, not their full cross-product, degree 200, or size 4,096.

Scale witness and confounding. Degree changes global K and CT preferences simultaneously. A posthoc typed probe keeps an edge endpoint's degree one and local list $\{0, 1\}$ fixed, while a separate star raises global Δ. Preferences Δ and $\Delta - 1$ make it the strict color-zero winner. All fifteen primary CF CT policies reject

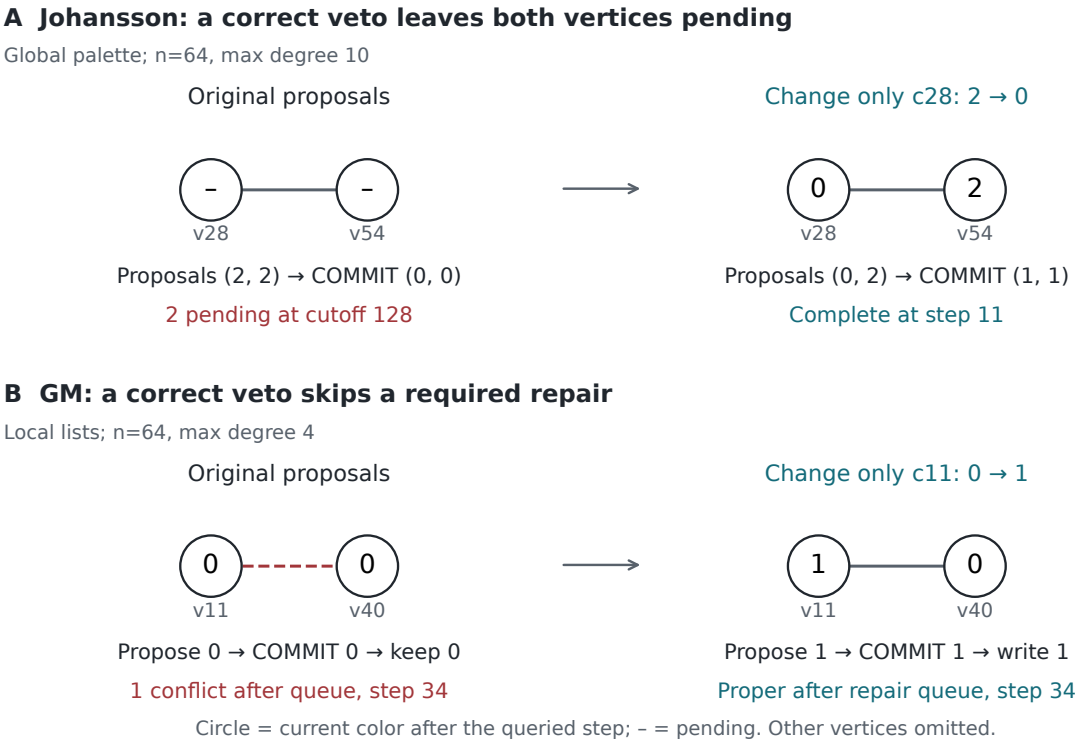


Figure 4: Two selected copied witness executions, showing local fragments after one queried application. A: Johansson steps 10 to 11, with free colors $\{0, 2\}$ at both vertices. B: GM steps 32 to 33, with list $\{0, 1\}$ at the scheduled vertex 11; the dashed edge is an existing conflict. Changing one proposal and recomputing learned COMMIT completes each copied run. Exact-COMMIT substitution on unchanged proposals preserves both failures. All other messages and first-stage tensors stay fixed at the intervention; these cases do not estimate a population rescue rate. Full outcomes are in Table 24.

its queried COMMIT from degree ten in the main embedding. Legacy preference/3 units alter responses but do not fix all tested cases. Both encodings preserve order. This is scale sensitivity of fixed policies, not an information obstruction, a demonstrated rounding cause or a random functional failure rate. No degree-only causal effect independent of K and private-input scale is identified.

8.1 Progress beyond correct confirmation

For fresh independent private draws, Equation (2) describes the exact anonymous rule's probability of writing. Proposition 5 makes the missing hypotheses explicit, including a bound on each neighbor's per-color offer probability. This covers the exact always-active Johansson variant, whereas coincident deterministic proposals can make progress zero (Counterexample 7). A finite draw probe demonstrates observed concentration, not exactly zero probability of an alternative color.

The GM suffix condition in Section 7 explains why a correct veto can leave an unresolved conflict without implying that every omitted repair must fail. The two candidate interventions distinguish these mechanisms in specific copied executions. The broader classification extends their observed scope without converting a retrospective replay into a guarantee over new graphs.

The finite-type MAX condition in Appendix F certifies one canonical COMMIT predicate when all 64 type subsets satisfy its sign inequalities. It does not certify candidate diversity, scheduled repair or palette-context invariance. All 45 selected canonical transfer cores fail that sufficient condition; five different historical E1 weights pass conditional certificates. Successful finite runs do not establish its hypotheses, and failure of a sufficient certificate is not itself a reachable failure. The identifier obstruction and representation controls remain separate from numerical sensitivity (Appendix D).

9 Discussion and Limits

Functional success, fidelity to a specified continuation and transfer answer different questions. The positive coverage establishes that these local policies can execute heterogeneous coloring procedures over substantial finite domains. It does not identify how a particular module responds to every alternative input. Transfer evaluates whether either property persists under a declared domain shift; good functional transfer does not imply continuation fidelity on unvisited queries.

The paired acquisition studies show that support and weighting determine what a continuation score measures. Native GM confirmations contain no negative color labels. BE's dominant negative type makes an occupancy-blind eligibility rule strong by occurrence BA, while equal-type accuracy rewards rarer decisions. Equal target counts do not equalize information. Nearly all coarse oracle configurations occur in CF training, despite disjoint fine input types. Neither new-logical-condition extrapolation nor an advantage over other informative-negative acquisitions is established. The wide BE seed-paired interval leaves its mean occurrence advantage uncertain; it does not establish a zero effect. These limits constrain the estimand, rather than invalidating every measured acquisition difference.

Transition relevance explains why these scores need not track completion. A silent root's confirmation cannot write; an active acceptance can also be inert when decoding preserves its old value. In contrast, missing a mandatory shrinking-bound write immediately breaks a KW invariant. The 30,000 fixed-component executions localize all selected CF failures: COMMIT-only replacements rescue 1,799, and participation-only replacements rescue the other 796. This demonstrates sufficiency of the respective component corrections on consumed cases, not a learned remedy or a general causal relation between fidelity and success. The new neighbor-context follow-up supplies a separate positive training intervention; it does not address every possible BE failure mechanism.

Johansson and GM show different limits of correct rejection. A veto cannot supply proposal diversity, and a repair veto must be judged against the remaining queue. A later endpoint may clear a conflict in some GM executions; the reconstructed witness has no such effective later repair. Conditional progress proofs state the independence, diffusion and coverage assumptions needed for guarantees. They are not certified properties of the networks.

Supplied scheduling, algebraic predicates and decoding delimit what is learned. Native packages use heterogeneous protocols, and no architecture ranking follows. Graphs, tapes and first-stage tensors are paired,

so decisions and conditions cannot be counted as independent training replicas. Degree transfer changes palettes and CT preference scale as well; the scale probe does not identify a degree-only effect. Degree 200 and size 4,096 pilots are exploratory and do not define a jointly tested operating region. Output-law nonrejection is not equality (Appendix J).

10 Conclusion

Local neural policies can achieve broad finite functional coverage and transfer while leaving specific continuation responses unresolved. Explicit contracts, support audits and component replays make those differences testable. In these models, better continuation scores sometimes coexist with lower completion because errors have different reachability and transition consequences. Proposal diversity and scheduled repair provide further mechanisms by which correct confirmation can coexist with failure. Learning a distributed rule should therefore be evaluated through declared functional protocols, continuation targets and transfer conditions together, with each result interpreted at its actual level of evidence.

A Paired Input-Context Study: Replicas and Costs

Table 10: All 40-model final seed outcomes. BA is a percentage; differences are percentage points. Successes are out of 1,000 per model. N expands native confirmations.

Teacher	Seed	N BA	CF BA	Difference	N success	CF success
BE	105001	99.435	52.756	-46.679	1000	100
BE	105002	99.334	99.872	+0.538	1000	1000
BE	105003	99.539	99.987	+0.448	1000	1000
BE	105004	50.193	99.923	+49.730	1000	1000
BE	105005	50.320	99.885	+49.565	1000	1000
BE	105006	99.152	99.897	+0.746	1000	1000
BE	105007	50.348	99.923	+49.575	1000	1000
BE	105008	99.195	52.094	-47.101	1000	101
BE	105009	99.462	99.859	+0.397	1000	1000
BE	105010	99.451	99.949	+0.498	204	204
GM	105001	50.000	100.000	+50.000	1000	1000
GM	105002	50.000	100.000	+50.000	1000	1000
GM	105003	50.000	100.000	+50.000	1000	1000
GM	105004	50.000	100.000	+50.000	1000	1000
GM	105005	50.000	100.000	+50.000	1000	1000
GM	105006	50.000	100.000	+50.000	1000	1000
GM	105007	50.000	100.000	+50.000	1000	1000
GM	105008	50.000	100.000	+50.000	1000	1000
GM	105009	50.000	100.000	+50.000	1000	1000
GM	105010	50.000	100.000	+50.000	1000	1000

Each pair physically acquires 10,200 optimization targets: one shared 3,000-target base and two 3,600-target auxiliary banks. Each model accesses 6,600 targets, so shared physical acquisition and summed model access differ. The shared auxiliary P/C context contains 7,200 root scalar occurrences and additional real-neighbor occurrences. Base root P/C context overlaps base target labels; its neighbor conditioning values are also recorded separately. These are scalar occurrences, not claims about distinct simulator calls.

`results/paper-revision-20261030/prospective-summary.json` records the per-seed component counts, complete-query duplicates, phase quotas, contexts considered/rejected, observed simulator states, acquisition CPU/wall time and paired bootstrap outputs. Training records separately retain presentations,

Teacher	Base target component	Count range across ten banks
BE	confirmation	944–944
BE	linial point	168–168
BE	participation	944–944
BE	proposal	944–944
GM	confirmation	744–754
GM	conflict mark	738–768
GM	participation	744–754
GM	proposal	744–754

Table 11: Base supervision sums to exactly 3,000 per seed. Both arms add 3,600 auxiliary COMMIT targets. Context values and simulator costs are reported separately, not absorbed into a free-label assumption.

visited targets, source/data hashes and optimizer CPU/wall time. One run's wall time includes a shared-host memory stall; wall time is not a FLOP or teacher-sample-efficiency estimate. All training remains at update 1,500. The original source/config freeze predates scientific training; the complete forty-weight freeze is dated 30 October 2026, 10:11:58 UTC, before any new final diagnostic generation. No final outcome selects a milestone, protocol change, seed exclusion or replacement model.

The recorded source freeze covers the enumerated files, not a complete transitive dependency or environment lock. In particular, graph generation, storage and sample-complexity data helpers outside that list are not sealed by this receipt. Their stability is not independently timestamped here. The original freeze is preserved; this limitation is not repaired by retrospectively adding hashes to it.

A.1 Prospective acquisition and sufficient key

This specifies the already executed acquisition, not a revised protocol. Color queries use four option indices, $q = c_v \in \{0, 1, 2, 3\}$, degree at most three, and legal teacher-prefix states. Auxiliary and diagnostic graph streams use separate namespaces; their sizes are respectively 8/12/16/24 and 28/36. The frozen configuration and graph generators identify the streams.

For each queried color, let w_j be permission and u_j the supplied pending-or-permitted bit in GM. In BE the queried-stage adapter sets $u_j = 1$. Let b_j be the current-own-color indicator for q in GM and zero in BE. Let o_{vj} be the neighbor's occupancy of the receiver's queried KW option in BE and b_j in GM. Set $\kappa_v = \operatorname{sgn}(a_v - q_{\mathrm{KW}})$ in KW, where $q_{\mathrm{KW}} = \lceil 4B/5 \rceil$, and zero in GM. Here a_v is the auxiliary color; it is distinct from the queried option index q. GM also supplies $\sigma_{vj} = \operatorname{sgn}(ID_j - ID_v)$ and the same-slot indicator τ_{vj}; both are zero in BE's queried-stage adapter. With the actual supplied messages p_j, c_j, define

$$r_v = (u_v, b_v, p_v, \mathbf{1}[u_v > 1/2]), \qquad m_{vj} = (u_j, o_{vj}, p_j, \mathbf{1}[u_j > 1/2 \ \wedge \ c_j = q], \sigma_{vj}, \tau_{vj}),$$

$$K = (T, \phi, w_v, \kappa_v, r_v, \operatorname{sort}\{m_{vj} : j \in N(v)\}).$$

The neighbor collection is a set: identical message types occur once. This is sufficient for the fixed-weight queried-slot MAX computation, not a minimal oracle quotient, a proposal key or a classifier of LINIAL/DETECT. The original code represents the BE offset as the signed encoded offset; its sign is exactly $\operatorname{sgn}(a_v - q_{\mathrm{KW}})$.

For the hash, all bits are integers 0/1 and signs are $-1, 0, 1$; T is the string `BE_BASE_D3` or `SPECULATIVE_GM`. The phase integer ϕ is 5 for KW, 1 for ASSIGN and 3 for REPAIR. Tuples/sets are serialized as JSON lists, with neighbor tuples in lexicographic order and separators `,` and `:` with no spaces. Prefix the UTF-8 JSON by the literal salt `revision-20261030-commit-types-v1`, with no separator. Interpret the first eight SHA256 bytes as a big-endian unsigned integer. Residue zero modulo five denotes held-out types. This reserves 20% of hash buckets; it need not reserve exactly 20% of observed types or occurrences.

Acquisition proceeds as follows.

1. Retain native base targets until the 3,000-component budget is filled. Color-phase base queries must have training keys; LINIAL/DETECT use their separate native targets and do not receive this COMMIT filter.

2. In each auxiliary color-phase batch with an unfilled quota, enumerate candidate roots with a running cursor i before any filtering. Set $k = i \bmod 4$ and $c'_v = (c_v + k) \bmod 4$. Draw an independent uniform value for every considered root. Set $p'_v = w_v$, except that $k = 3$ and a draw below $1/4$ set $p'_v = 0$. All real-neighbor P/C values stay native. Advance the cursor for all considered roots, including rejected roots.

3. Training retains a root only when both $K(S, p, c)$ and $K(S, p', c')$ are training types, subject to the 3,600 KW or 1,800 ASSIGN/1,800 REPAIR quotas. Compute its labels only after this filter. The native arm receives one native confirmation and CF one intervened confirmation on the same state.

4. After freezing all forty models, the diagnostic stream applies the same intervention, but retains a root if its intervened key is held out. The native key is unrestricted at test. Retain exactly 4,000 KW or 2,000 ASSIGN/2,000 REPAIR queries per seed, without filtering on truth labels.

The intervention RNG is seeded by the frozen namespace, teacher, training seed, acquisition kind and the literal `intervention`. Native graph/tape streams have their separate base/auxiliary/diagnostic suffixes. Filtering can change offset and silence frequencies; the pre-filter silence probability is not a claim about the retained distribution. No phase quota balances the positive and negative truth labels within that phase.

A.2 Post-test coverage and confusion audit

All tables here are descriptive additions after the final test. Labels are recomputed from raw encoded state arrays and agree with stored base, auxiliary and final color confirmations. Fine keys are reconstructed from those arrays; their training/test unions equal the delivered key files and obey the original hash filter. Model predictions are summarized from frozen result records; this original array audit does not rerun neural inference or train another model. A separate complete checkpoint re-inference reproduces all final diagnostic metrics for all 40 models (160,000 decisions), using the original phase/batch context. This verifies the recorded predictions without changing the post-test status of these additional endpoints.

Teacher	Native logical overlap (%)	CF logical overlap (%)
BE	2.125–2.900	99.475–100.000
GM	37.725–39.500	100.000–100.000

Table 12: Both acquisitions' coarse oracle-configuration overlap with the same test, range over ten seeds. All fine training/test keys remain disjoint.

Table 13: Ranges over ten seeds of base plus auxiliary color-COMMIT targets. LINIAL and DETECT have different targets and are excluded. Minima and maxima in separate columns need not occur at the same seed.

Teacher	Phase	Arm	Positive labels	Negative labels
BE	KW	native	152–189	4,355–4,392
BE	KW	cf	113–147	4,397–4,431
GM	ASSIGN	native	2,417–2,440	0
GM	ASSIGN	cf	2,108–2,153	277–326
GM	REPAIR	native	1,904–1,932	0
GM	REPAIR	cf	1,119–1,171	749–793

Table 14: Final support and programmed diagnostic by seed. Types are distinct fine sufficient keys; mass is the largest key's fraction of 4,000 queries. Overlap is the fraction of occurrences whose coarser oracle configuration was seen by CF training. Mass, overlap, BA and type accuracy are percentages.

Rule	Seed	Types	Max mass	Overlap	Gate BA	Gate type acc.
BE	105001	20	95.38	99.750	99.679	80.000
BE	105002	23	96.03	99.475	99.770	78.261
BE	105003	17	96.30	99.825	99.770	82.353
BE	105004	20	95.17	99.750	99.614	80.000
BE	105005	18	95.62	100.000	99.731	83.333
BE	105006	20	95.05	99.800	99.589	80.000
BE	105007	21	95.15	99.725	99.704	80.952
BE	105008	20	95.53	100.000	99.706	85.000
BE	105009	22	95.67	99.725	99.756	86.364
BE	105010	17	96.23	99.775	99.757	76.471
GM	105001	89	23.80	100.000	53.055	61.798
GM	105002	84	23.62	100.000	52.910	63.095
GM	105003	92	23.82	100.000	53.204	63.043
GM	105004	86	24.60	100.000	53.182	62.791
GM	105005	91	23.88	100.000	52.686	63.736
GM	105006	83	24.10	100.000	52.921	60.241
GM	105007	86	24.52	100.000	53.193	62.791
GM	105008	87	23.85	100.000	53.111	60.920
GM	105009	89	23.80	100.000	53.065	64.045
GM	105010	82	24.60	100.000	53.005	60.976

Table 15: Confusion counts summed over ten final banks, 40,000 queries per teacher/procedure. Within eligible active queries, positives are free options and negatives occupied options. Silent and ineligible queries are negative. No GM query is ineligible. These are accounting totals, not IID replications.

Rule	Stratum	Procedure	TP	TN	FP	FN
BE	eligible active	native	995	0	228	0
BE	eligible active	cf	814	228	0	181
BE	eligible active	gate	995	0	228	0
BE	eligible silent	native	0	91	42	0
BE	eligible silent	cf	0	52	81	0
BE	eligible silent	gate	0	133	0	0
BE	ineligible	native	0	26,956	11,688	0
BE	ineligible	cf	0	38,644	0	0
BE	ineligible	gate	0	38,644	0	0
GM	eligible active	native	15,554	0	22,963	0
GM	eligible active	cf	15,554	22,963	0	0
GM	eligible active	gate	15,554	0	22,963	0
GM	eligible silent	native	0	0	1,483	0
GM	eligible silent	cf	0	1,483	0	0
GM	eligible silent	gate	0	1,483	0	0

The reproducible analysis and input hashes are `scripts/audit-revision-continuations.py` and `results/paper-revision-20261030/continuation-audit.json`.

A.3 Selected BE failure diagnoses after the test

Frozen-model inference covers both arms of BE seeds 105001, 105002, 105004, 105008 and 105010, plus both GM arms in 105001. This deliberately selected set includes good and bad outcomes; it does not estimate their prevalence. There are 102,128 COMMIT predictions on existing native-color, auxiliary-training and final banks. All 48,000 final predictions reproduce the original confusion counts; no weights, split or primary results are changed. This diagnosis uses neural inference, separately from the preceding array-only coverage audit.

CF seed	Aux. BA	Final BA	Aux. FN	Final FN	Final FP
105001	99.510	52.756	1	98	10
105008	100.000	52.094	0	83	16

Table 16: Selected BE COMMIT generalization failures. BA is a percentage. Each row evaluates the arm's 3,600 acquired auxiliary queries and the 4,000 held-out final queries. All auxiliary targets were visited during training; neither selected model has an auxiliary false positive. The contexts differ, so this is a descriptive fit/test contrast, not a new randomized comparison.

The two CF models also classify all 944 native-color base queries correctly. Their held-out-context errors coexist with near-perfect acquired-label fit; this does not identify a unique shortcut, optimization cause or limitation of representational capacity.

For seed 105010, native and CF first-stage tensors are identical. On 4,000 teacher-prefix test states their participation head omits 13 of 118 eligible roots, with no false activation. On 3,600 auxiliary states it omits 4 of 128 eligible roots and falsely activates 1 ineligible root. Auxiliary P/C values condition the COMMIT loss; they are not direct participation-loss targets. These first-stage scores on auxiliary states are evaluations, not evidence of failure to fit a participation training objective there. Candidate options equal the native ones throughout both banks, with no occupied proposals at eligible roots.

The witness is `final-s105010-0000`, a 32-node path in the native arm. Replay preserves the first 32 original cases in their original order, cohort size 32 and root batch size 1,024. All 32 terminal color vectors, step counts, failures, invariant flags and correctness flags match the original records. In the selected case the state before step 7 is a proper auxiliary coloring bounded by 21. Vertex 31 has value 17 and must recolor when the interpreter lowers the bound to 17. Its participation logit is -0.420059, so it waits, leaving 17 outside $[0, 17)$. Its option 0 decodes to the free value 12, also chosen by the exact native rule. The actual COMMIT logit -11.225789 correctly rejects silence. Setting only this root's participation bit to true, with identical first-stage tensors, candidates and all other messages, gives logit +4.629150 and the correct positive oracle label. The observer's re-evaluated actions match the original actions at this frame. The intervention is only a local query; unchanged actions drive the replay. It locates the first violation in this case, without showing that all 796 failed runs share this mechanism or that the intervened run would complete. The subsequent full-battery interventions in Section 7.1 separately establish that participation-only correction rescues all 796 failures in each paired arm; that stronger result is not inferred from this earlier local query alone.

`scripts/probe-revision-be-failures.py` reproduces the diagnosis; `results/paper-revision-20261030/be-failure-diagnostics.json` records input hashes, counts, graph/tape seeds and the local intervention. This is post-test explanatory evidence, not independent scientific review.

B Component Replays and Additional Evidence

All baselines reproduce complete stored terminal fields in their original cohort 32/inference-batch 1,024 float32 CPU context. The six checkpoints are chosen because their three seeds include every functional failure in the input-context study; the two nonfailing native arms are paired controls. No checkpoint is retrained, no failing case removed and no new primary endpoint introduced. Baseline, COMMIT-only, participation-only, candidate-only and all-KW modes are fixed before outcomes. All six models and all cases remain in the output, including ineffective interventions. The selected CF failure count is 2,595, which is also the complete CF failure inventory for that study's original 10,000 cases. Rescue is a statement about those cases, not a population success probability.

Seed	Arm	Learned	Exact CMT	Exact P	Exact CAND	Exact KW
105001	Native	1000	1000	1000	1000	1000
105001	CF	100	1000	100	100	1000
105008	Native	1000	1000	1000	1000	1000
105008	CF	101	1000	101	101	1000
105010	Native	204	204	1000	204	1000
105010	CF	204	204	1000	204	1000

Table 17: All six selected BE models on their original 1,000-case batteries. CMT replaces only confirmation, P only participation, CAND only candidate; the dependent learned confirmation is recomputed when messages change. Exact KW replaces all three. Linial remains learned. These are 30,000 diagnostic executions on consumed cases, not 30 training replicas.

Contract rejection need not mean an invalid write. The baseline diagnostic records 64 and 20 active acceptances rejected by the contract in native seeds 105001 and 105010. A subsequent unchanged-policy observer reproduces both 1,000-case batteries exactly and checks the actual KW decoding. Every one of these 84 writes reproduces its old auxiliary color and satisfies the new bound. The contexts (B, a, c, D) are $(8, 6, 2, 6)$, $(12, 9, 1, 9)$ and $(14, 11, 3, 11)$, with respective counts 53/15, 6/1 and 5/4 in the two seeds. They are errors against the specified contract but do not alter that transition, as Proposition 2 predicts. The frozen counter named `learned_unsafe_accepts` in this BE diagnostic means active contract-rejected acceptances, not observed collisions; it must not be conflated with transfer's forbidden-write counter. This observer is posthoc and does not revise the frozen intervention outputs.

Campaigns and label budgets. The historical diagnostic uses five seeds, unequal acquisition budgets and development-reused queries. The 40-model input-context study uses ten paired seeds, a shared 3,000-target base, 3,600 auxiliary targets per arm, 1,500 fixed updates and a fine-key hash holdout. The 100-model label-budget study uses five paired seeds, 4,000 updates and different intervention families, support and endpoints; its 1,500-update snapshots are secondary. The five-model neighbor-message follow-up freezes its modified acquisition after inspecting that study and uses a fresh v2 evaluation battery. The studies are not pooled and their equal-type metrics differ: the former averages accuracy equally over types, while the latter also class-balances its type endpoint. Numerical bounds used by a generator are not necessarily observed by the COMMIT head.

The label-budget study has 32/40 fully passing BE checkpoints; only two of five high-CF seeds meet the 99% unseen-type criterion. All acquisition variants are retained, including the no-force and late-bound conditions. Complete component-group accounting gives slightly different native/CF label totals; neither identical logical support nor equivalent class information follows from a matched nominal budget.

What further experiments would identify. Attributing an advantage specifically to counterfactual label generation would require a control with informative positive and negative labels, matched support/balancing, updates and checkpoint selection. Generalization to unseen interventions would require holding out complete families and checking novelty at the effective model input, rather than only at a generator parameter. Neither stronger claim is made here.

C Native Coverage: Protocol and Uncertainty

The 21/21 result and its cohort table are in Section 5. The following details delimit its finite statistical interpretation.

Final tests use sizes 32, 48, 64; family weights differ: seven historical strata versus ten new strata of 100 runs. The 105 selected policies account for 105,000 successful observations, not a success probability of an average policy. Five training seeds are the treatment replication units. Under independent Bernoulli trials with a common probability, 1,000/1,000 gives pointwise two-sided 95% Clopper–Pearson lower limit (Clopper and Pearson, 1934) $0.025^{1/1000} = 0.996318$; the one-sided limit is $0.05^{1/1000} = 0.997009$. For fixed independent

nonidentical strata, AM–GM makes the zero-failure formula conservative for the weighted average, not each stratum. No simultaneous all-policy guarantee or universal correctness follows. A selected successful budget is not a theoretical minimum. The run-level interpretation also assumes semantically neutral batching. Real-arithmetic message separation supports that premise; universal numerical neutrality across floating-point batch contexts has not been certified.

Preserved negative results. The historical study's nine noncertifications under 500 updates are not impossibility results. The pre-revision native-certification study retains 21 failed executions among 195,000 final observations across 39 packages and 170 distinct checkpoints; some weights receive new batteries. All failures remain separate from the selected all-pass set. Independent fixed-bank E1 controls compare 500 versus 1,500 updates along the same seed/checkpoint lineage, holding labels fixed and using fresh final tapes. This is a more narrowly identified optimizer contrast than matched versus CF; it does not establish a CF effect or monotone benefit from more updates (Appendix G).

D Information and Identifiers

Consider an initial two-node JP graph with identical anonymous observations and fixed private inputs. Swapping the distinct IDs changes the required root action from write to wait without changing the anonymous observation.

Proposition 3 (Observation indistinguishability)**.** *If two admissible contexts have the same observation and different deterministic target actions, no deterministic function of that observation is correct on both. A randomized function cannot be almost surely correct on both.*

Proof. Equal inputs give equal outputs or equal conditional output laws. Different point-mass targets cannot both be realized. For equiprobable binary contexts, average correctness is at most 1/2. □

This elementary interface obstruction is not a new distributed lower bound. It requires indistinguishability, not numerical closeness, and does not preclude solving coloring by a different rule. Historical integrated JP/GM already had numeric ID differences; their failures were not ID blindness.

JP representation	94101	94102	94103	94104	94105
Without ids	6/60	6/60	6/60	6/60	6/60
Numeric ids	13/60	11/60	6/60	26/60	15/60
Numeric differences	16/60	15/60	19/60	31/60	20/60
Relative comparisons	60/60	60/60	60/60	60/60	60/60

Table 18: Exploratory E1 JP control: same dimensions, initialization, 3,000 labels and 500 updates. Entries are development successes out of 60, not final 1,000-run certifications.

Relative comparisons achieve 60/60 in all five seeds. This supports representation sensitivity under this budget. Numeric IDs/differences retain order information, so finite failures do not prove inability to learn it. Signs remove the need to resolve gaps as small as 1/65536; they must arise from one coherent order. BE uses actual ID arithmetic: increasing numeric transformations need not preserve its polynomial trajectory.

E Exact Rules and Conditional Guarantees

Assume finite simple graphs, proper partial states where required, declared lists of size at least $d(v)+1$, coherent observations and the specified synchronous exchanges/schedules. Code domain guards and independent scalar tests complement these arguments.

Lemma 4 (Availability)**.** *For a pending vertex,* $|\mathcal{A}_v| \geq |C_v| - (d(v) - d_U(v)) \geq d_U(v) + 1$.

Proof. At most one distinct color per fixed neighbor is forbidden. Counting duplicates only weakens the bound; differing local lists cause no problem. □

Luby/Johansson. Proposals avoid fixed colors, and exact equal-active-offer rejection prevents simultaneous collisions. Permanent colors persist, giving safety by induction. Condition on history/state before new private inputs, not a snapshot already containing those inputs. If $q_v(c)$ is the proposal law, p_v its independent participation, with independent pairs across nodes, then

$$\Pr(v \text{ commits} \mid S) = p_v \sum_{c \in \mathcal{A}_v} q_v(c) \prod_{u \in U_v} (1 - p_u q_u(c)). \tag{2}$$

For uniform proposals and $p_u = 1/2$, the chord bound $\log(1-x) \geq -2\log(2)x$ on $[0, 1/2]$ and Jensen yield $\frac{1}{2}2^{-d_U(v)/|\mathcal{A}_v|} \geq 1/4$. For always-active Johansson, each pending neighbor of v has at least two free colors, so $q_u(c) \leq 1/2$. The same argument yields $2^{-2d_U(v)/|\mathcal{A}_v|} \geq 1/4$. These prove the specified variants' bounds; they do not import a different algorithm's theorem (Luby, 1993; Johansson, 1999).

ColorTrials. Let $D \geq \Delta$ be the common preference bound: $D = 3$ natively, even on lower-degree graphs, and $D = \Delta$ in the extension. Pending nodes draw independent preferences for each color in $\{0, \ldots, D\}$. Only pending neighbors for which the color is available compete. The minimum strictly winning available color is chosen; otherwise wait. Adjacent vertices cannot both strictly win one color. For any $c \in \mathcal{A}_v$, the event of preference D at v and smaller preferences at all at most Δ competitors has probability at least $\rho_{\mathrm{CT}} = (D+1)^{-1}(D/(D+1))^{\Delta} > 0$ for $D > 0$. It guarantees some write at v, perhaps a lower winning color. Isolates at $\Delta = 0$ finish in one step. The common global preference alphabet matters; different local alphabets need another analysis. This is a repeated subroutine inspired by symmetry breaking (Schneider and Wattenhofer, 2010), not a full multi-phase complexity claim.

Random termination. If each pending vertex has conditional progress at least $\rho > 0$ at every reachable state, iterated conditioning gives $\Pr(T_v > t) \leq (1-\rho)^t$ and $\Pr(T > t) \leq \min\{1, n(1-\rho)^t\}$. Thus ideal exact rules terminate almost surely. Transfer uses at least 128 steps, enlarged for CT until this ideal tail bound is at most 10^{-9}. Neither arbitrary learned policies nor finite PRNG orbits inherit the result automatically.

JP and Greedy. JP's pending strict maxima of coherent distinct IDs form an independent set. First-fit exists by availability, and the remaining global maximum writes each step, giving safety and at most n steps. This synchronous variant is inspired by Jones and Plassmann (1993). Greedy permits exactly one vertex in increasing-ID order per step; first-fit avoids prior colors and completes in n writes. Global scheduling cost is supplied, not a free LOCAL operation.

Speculative GM. Split increasing-ID order into $\min(4, n)$ balanced blocks. ASSIGN visits one position per block simultaneously. A later assignment avoids all earlier neighbor colors, so only equal-slot vertices can conflict. DETECT marks the smaller-ID endpoint of each equal-color/equal-slot edge. Every conflict is covered; the unmarked subgraph is proper. Serial repair visits marked vertices once in increasing-ID order and avoids all current neighbor colors. The last repaired endpoint of any edge avoids the other; edges with no repaired endpoint were already proper. Correctness follows in $2\lceil n/\min(4,n)\rceil + m$ steps, $m \leq n$, under these sufficient assignment/mark/free-write hypotheses. A missed learned repair receives no retry at that vertex; a later endpoint may still clear its edge. This is the implemented four-block variant (Gebremedhin and Manne, 2000), including supplied barriers/queue, not a guarantee for arbitrary learned marks.

BE degree-three base case. Distinct IDs are initial proper auxiliary colors. Degree-at-most-four polynomials over $\mathbb{F}_{13}$ encode them. A different neighboring polynomial agrees at at most four points, so three neighbors block at most twelve and $13 > 3 \cdot 4$ leaves a point. Encode selected point/value below $13^2 = 169$. A second degree-at-most-two reduction over $\mathbb{F}_7$ has a point because $7 > 3 \cdot 2$, yielding bound 49. Different chosen points give different pairs; equal points have different unblocked values, proving properness (Linial, 1992). For bound $B > 4$ set $q = \lceil 4B/5 \rceil$. Colors below q remain; each high old color owns a disjoint four-option block in $[0, q)$, possible since $4(B-q) \leq q$. Three neighbors forbid at most three options. High–high neighbors have disjoint blocks; high–low collisions are excluded by free options. The sequence $49 \to 40 \to 32 \to 26 \to 21 \to 17 \to 14 \to 12 \to 10 \to 8 \to 7 \to 6 \to 5 \to 4$ has thirteen KW reductions (Kuhn and Wattenhofer, 2006). Decoding does not search for a good point/option. This is only the degree-three

base construction associated with Barenboim and Elkin (2008), not the general recursion. At degree four, $13 > 4 \cdot 4$ fails; changing constants changes the variant.

F Conditional Representation and Probability Results

The following elementary arguments are not claimed new general GNN-learning theorems. Checkpoints satisfy their hypotheses only where verified.

Proposition 5 (Diffuse proposals and bounded neighbor offers)**.** *Assume a proper partial state, exact confirmation, independent node-private participation/candidate pairs, independence within each pair, and proposal support in $\mathcal{A}_u$. For a pending root v, suppose $q_v(c) \le \beta/|\mathcal{A}_v|$ and $p_u q_u(c) \le \theta < 1$ for all $u \in U_v, c \in \mathcal{A}_v$, with $\theta > 0$. Then*

$$\Pr(v \text{ commits} \mid S) \ge p_v(1-\theta)^{\beta \sum_{u \in U_v} p_u/(\theta|\mathcal{A}_v|)}.$$

If $0 < p_{\min} \le p_u \le p_{\max} < 1$, setting $\theta = p_{\max}$ gives $p_v(1-p_{\max})^{\beta d_U(v)/|\mathcal{A}_v|} \ge p_{\min}(1-p_{\max})^{\beta}$. For always-active uniform Johansson, $\theta = 1/2, \beta = 1$ gives at least $2^{-2d_U(v)/|\mathcal{A}_v|} \ge 1/4$.

Proof. The chord bound is $\log(1-x) \ge ax$ for $a = \log(1-\theta)/\theta < 0$ and $0 \le x \le \theta$. Set $\Lambda_c = \sum_{u \in U_v} p_u q_u(c)$. Equation (2) is at least $p_v \sum_c q_v(c) e^{a\Lambda_c}$, which by Jensen is at least $p_v e^{a \sum_c q_v(c)\Lambda_c}$. The weighted load is at most $\beta \sum_u p_u/|\mathcal{A}_v|$, since each neighbor has at most unit mass on $\mathcal{A}_v$. Multiplication by $a < 0$ reverses that inequality, proving the first bound. Availability and $\sum_u p_u \le p_{\max} d_U(v)$ give the second. For exact Johansson, a pending neighbor has at least two available colors, so $q_u(c) \le 1/2$. Isolates use the empty product and write with probability p_v. These bounds are conditional sufficient conditions, not properties established uniformly for learned checkpoints. □

Corollary 6 (Diffuse submass)**.** *In the $p_{\max} < 1$ case, if the concentration bound holds only on $S_v \subseteq \mathcal{A}_v$ of mass $\eta_v > 0$, the lower bound is $p_v \eta_v (1-p_{\max})^{\beta d_U(v)/(\eta_v|\mathcal{A}_v|)}$. Uniform $\eta_v \ge \eta > 0$ at every reachable state implies uniform progress $p_{\min}\eta(1-p_{\max})^{\beta/\eta}$.*

Proof. Drop nonnegative terms outside S_v, renormalize by η_v, and repeat. $xb^{\beta/x}$ increases for $x > 0$, $0 < b < 1$, since its derivative is $b^{\beta/x}(1 - \beta \log b/x) > 0$. □

Counterexample 7 (Safety without useful progress)**.** *On a K-clique with $K \ge 2$ colors, deterministic color-zero proposals, participation $1/2$ and exact rejection progress only with exactly one active vertex: probability $K/2^K$, expected first-progress time $2^K/K$. Always-active vertices never progress. A shared uniform candidate draw has uniform individual marginals and the same slow law; independence in (2) cannot be replaced by marginal uniformity.*

Proposition 8 (Canonical finite-type MAX COMMIT)**.** *Assume legal canonical per-color inputs/gating, nonnegative shared second messages, componentwise MAX with zero empty aggregate, and exact proposed-slot gather. The pending active center's queried-slot vector is $(1,0,1,1)$; neighbor vectors $(\text{pending}, \text{owns}_c, \text{participates}, \text{offers}_c)$ are among*

$$(0,0,0,0), (0,1,0,0), (1,0,0,0), (1,0,0,1), (1,0,1,0), (1,0,1,1).$$

If all 64 type-subset COMMIT signs accept precisely subsets without the fixed-color or active-equal-offer conflict types, the rule is exact at any legal degree/palette in real arithmetic and preserves properness.

Proof. MAX is associative, commutative and idempotent; multiplicities disappear. Nonnegative messages admit zero identity. Gathering the proposed slot removes other slots. The 64 subsets therefore enumerate all aggregates. Excluding the two conflict types prevents old/simultaneous collisions. Induction gives safety, with no progress conclusion. □

This does not cover continuous preference or native residual contexts, multiple offers, SUM or MEAN. Outward-rounded float64 intervals enclose float32-stored weights as real coefficients. A separate mode adds γ_{m+2} float32 affine error and a subnormal allowance, conditional on IEEE rounding, no overflow and no

TF32/lower precision. It is not a proof for arbitrary GPU kernels. Five historical E1 weights pass; all 45 selected Luby/Johansson/Greedy transfer cores fail the sufficient all-types condition. Failure of a sufficient certificate alone does not prove a reachable failure.

Proposition 9 (MEAN dilution; SUM construction)**.** *Fix degree/palette-independent messages m_o, m_f and a continuous classifier F for a fixed active center. If $F(m_o) > 0$, a MEAN interface accepts a conflicting star with sufficiently many ordinary neighbors. A SUM interface can represent exact conflict presence in real arithmetic.*

Proof. A star with d ordinary leaves and one fixed neighbor owning the candidate has mean $(dm_o + m_f)/(d+1) \to m_o$. Continuity and strict acceptance imply false acceptance at some finite d. If F is L-Lipschitz, it suffices that $d+1 > L\|m_f - m_o\|_\infty / F(m_o)$. For SUM, use the legal-type conflict bit $b = o + \mathrm{ReLU}(p + a - 1)$ and head $1/2 - \sum_u b_u$. □

The MEAN statement assumes no separate degree/palette input and says nothing about reachability of the constructed state. SUM's observed failure is not impossibility of representing this predicate. An order-free equivariant slot scorer with identical available-slot inputs and invariant pooled context gives tied scores for fixed private input; it cannot select distinct ordered ranks without order features or external tie-breaking. Main scorers have positions, so this obstruction does not explain their errors automatically.

G Training, Certification and Historical Provenance

Table 19: Selected native packages: actual acquired labels per seed (ranges where necessary), updates and total parameters. Every row has five policies and 1,000/1,000 successes each. Historical seeds 83101–83105, new 94101–94105.

Teacher	Core	Cohort	Labels	Updates	Parameters	Pass
Luby	E1	Historical	999	500	16035	5/5
Luby	DeepSets	Historical	3000	500	17907	5/5
Luby	Transformer	Historical	999	500	16995	5/5
Johansson	E1	Historical	999	500	16035	5/5
Johansson	DeepSets	Historical	999	500	17907	5/5
Johansson	Transformer	Historical	999	500	16995	5/5
ColorTrials	E1	New CF	70000	4000	19396	5/5
ColorTrials	DeepSets	New CF	70000	4000	21268	5/5
ColorTrials	Transformer	New CF	70000	4000	20356	5/5
JP	E1	Historical	999	500	19396	5/5
JP	DeepSets	Historical	999	500	21268	5/5
JP	Transformer	Historical	999	500	20356	5/5
BE base	E1	New CF	6776	4000	19396	5/5
BE base	DeepSets	New CF	6776	4000	21268	5/5
BE base	Transformer	New CF	6776	4000	20356	5/5
GM	E1	New CF	6104–6120	500	19396	5/5
GM	DeepSets	New CF	6104–6120	500	21268	5/5
GM	Transformer	New CF	6104–6120	500	20356	5/5
Greedy	E1	Historical	999	500	16035	5/5
Greedy	DeepSets	Historical	999	500	17907	5/5
Greedy	Transformer	Historical	999	500	16995	5/5

Historical sources are `results/sample-complexity/certification.json`, frozen candidates and raw final rows. New sources are `results/multiteacher-counterfactual/plans/successful-policies.json` and confirmation freezes/rows. Prefix budgets reuse labels and seeds; they are not independent replications

or theoretical sample-complexity scaling. The new packages do not rewrite historical noncertifications as successes.

Source commits include Pedro audit `6836d855`, paper study `b5427faf`, integration `a54a18e3`, sample complexity `61f39760`, counterfactual `291d4b82`, degree `71153993`. These are branches/worktrees in shared history, not independent copies of every experiment. Old rows/protocols remain immutable; the previous manuscript is archived under `paper/ucgnn/archive/pre-closure-71153993.tex/pdf`.

Optimizer controls and costs. Independent E1 confirmation compares 500/1,500 updates at nominal 3,000 native labels for Luby, JP and BE, five paired seeds and new final tapes. Only optimizer effort differs along the same training lineage. Six arms and all seed outcomes remain in `results/sample-complexity/optimization-confirmation/results.json`. It identifies additional-optimization effects for those bank/seed conditions, not CF effects or optimal/monotone update counts. Luby and JP complete 1,000/1,000 per seed at both milestones. BE moves from 72,90,84,92,42 successes per 1,000 to 1,000 in every seed; these are the strongest resource-controlled optimizer results in the present evidence. The small-width and no-communication baselines below share native prefixes, 500 updates and paired seed conventions with E1. They are development controls, not optimized competitors or proof that every coloring rule needs communication. No-communication erases received states, relations and masks while retaining local inputs and external permissions. Small uses widths 8/4. Dataset manifests record

Teacher	E1	Small	No communication
Luby	28, 28, 28, 28, 28	0, 0, 28, 1, 0	0, 0, 0, 0, 0
JP	28, 28, 28, 28, 28	10, 0, 15, 0, 0	0, 0, 0, 0, 0
BE base	8, 5, 5, 6, 3	0, 0, 0, 0, 0	0, 0, 0, 0, 0

Table 20: Native development controls at nominal 3,000 labels and 500 updates. Each entry lists successes out of 28 for seeds 83101–83105 individually.

simulation steps, native/added labels, generation CPU/wall time and bytes. In the historical CF acquisition, four labels per color root are charged even if queries coincide: these are conservative acquisition counts, not distinct input counts. Checkpoints retain Adam, all RNG states, visitation and query caches. The archived closure step itself trained no scientific models; the separate forty-model prospective trial has its own training/freeze receipts. Disposable technical gradient tests produce no weights used in results. The evidence ledger identifies all selected hashes and full label-generation costs.

Teacher	Native labels	Added CF labels	Bank CPU seconds	Stored MB
BE base	3000	3776	0.435–0.592	0.874–0.876
ColorTrials	30000	40000	2.153–2.215	1.172–1.186
GM	3000	3104–3120	0.857–0.929	2.002–2.091

Table 21: Acquisition cost of the complete shared native/CF bank, range over five seeds. CPU time includes simulation, encoding and both label types; it is not the isolated marginal CF cost. Decimal MB includes replay witnesses. Training costs are separate.

H Continuation Diagnostics and Dependence

The sufficient key comes from actual BE second-stage inputs, not its label. Real centers have pending bit one and own-slot bits zero; ID/slot residuals are zero. KW substitutes receiver-relative option equality for neighbor own bits. QueryPolicy signs the offset. The root queried slot has offer bit one; other slots vanish under gather. Each valid neighbor contributes occupied query bit, participation and offer/query equality; MAX removes duplicates. With phase, root permission, signed offset and root participation, this determines the real queried-slot logit at fixed weights. It is not a sufficient key for proposals. Train keys include native and all stored CF queries. There are 58–65 distinct training keys and 115–132 diagnostic keys per seed, with

	Correct / 99,840				Unseen-type correct		
Seed	Passive	Matched	CF	Subset size	Passive	Matched	CF
94101	41,591	9,343	99,840	549	229	158	549
94102	9,157	9,322	99,818	459	270	342	437
94103	9,146	9,146	99,840	717	229	229	717
94104	42,262	9,076	99,828	522	288	182	510
94105	10,865	9,382	99,840	722	356	246	722

Table 22: Exact BE/E1 original numerators and posthoc novelty subsets. Subset sizes count correlated query occurrences, not distinct types.

highly unequal occurrence counts. Complete serialized input hashes retain irrelevant fields and measure a different novelty notion.

The diagnostic reconstruction is `scripts/final-counterfactual-audit.py`. It reuses the original queries/RNG, reproduces all numerators, and records each checkpoint and bank hash. Stored label checks find zero inconsistencies; they validate the stated oracle, not graph-wide realizability of every query. We use five paired seed ranges, not decision-level significance. Even five same-sign pairs give exact two-sided sign-test $p = 2/2^5 = 0.0625$ under an exchangeable-sign null, before multiplicity. Shared BE/GM heads and repeated states cannot increase the independent training sample size. Positive counts across seeds are 3,070,2,895,2,891,2,773,3,115; full confusion/eligibility strata are in `results/final-closure/review-ml-be-strata.json`. For unseen-type subsets positives are 52,108,90,56,60 and negatives 497,351,627,466,662. All arms have zero false negatives. The distinction concerns rejecting negative continuations under the explicitly chosen oracle; native traces alone do not uniquely specify off-policy semantics.

I Failure Inventory and Statistical Units

The complete table is `results/final-closure/generalization-cells.csv`. The factor table separates teacher, architecture, checkpoint, seed, size, degree, palette, family and cohort within stage. The following lists all failed confirmatory cells, each with 1,000 runs and $n = 64$. `failed-cells.json` and

Teacher/core	Cohort	Seed	Δ	Successes	Palette
Johansson/Transformer	Historical primary	83101	10	972	Global
GM/Transformer	CF primary	94102	4	819	Local
CT/E1	Native matched	94101	3	999	Global
CT/E1	Native matched	94101	4	999	Global
CT/E1	Native matched	94101	4	999	Local
CT/E1	Native matched	94105	3	994	Global
CT/E1	Native matched	94105	4	987	Global
CT/E1	Native matched	94105	10	912	Global
CT/E1	Native matched	94105	4	993	Local
GM/Transformer	Native matched	94102	4	819	Local
GM/Transformer	Native passive	94102	4	819	Local
CT/E1	Equal-update matched	94104	3	864	Global
CT/E1	Equal-update matched	94104	4	855	Global
CT/E1	Equal-update matched	94104	10	547	Global
CT/E1	Equal-update matched	94104	4	835	Local

Table 23: All fifteen failed cells. CT denotes ColorTrials; equal-update controls are not retrospectively assumed native certified.

`failure-replay-audit.json` give full checkpoint/row hashes, graph/tape seeds, first relevant differences and trace paths. All fifteen first failures reproduce with original batch context. Terminal auditing reconstructs

graphs/fingerprints, conflicts, incompleteness and palette violations and checks stored invariant/failure fields. It does not pretend to reconstruct unrecorded intermediate states for all runs.

Graphs/tapes repeat across cores, teachers, arms and global/local lists. Each checkpoint's instance-index battery supports its own pointwise inference; pooling dependent observations across policies does not. Realized family counts vary under iid uniform family selection. No aggregate defines a sampled population of trained algorithms.

I.1 Complete outcomes of the selected component interventions

Witness	Intervention	Steps	Terminal outcome
Johansson	baseline	128	cutoff
Johansson	candidate once	11	complete/proper
Johansson	exact commit once	128	cutoff
GM	baseline	34	collision
GM	candidate once	34	complete/proper
GM	exact commit once	34	collision

Table 24: Posthoc component interventions in two selected first-failure witnesses, not new population success rates. Baselines preserve the original 16-case inference cohort. Exact confirmation is evaluated on the actual learned messages.

The source is `results/paper-revision-20261030/component-rescues.json`. Figure 4 extracts actual vertices, colors and proposals from those records and the original witness frames. The full 64-node graphs, case/tape identities and unchanged 16-case cohort remain part of the diagnosis; the two-node drawing omits the rest of each graph.

J Output Laws and Preserved Distribution Studies

For fixed labeled G, let $\mu_{\pi,G}$ be the full terminal-vector/status law including failures/cutoffs. Labels, palette, interpreter, cutoff, conditioning and color-name quotient must agree before comparing studies. Equal vertex marginals do not determine a joint law: equally weighted $(0,0)/(1,1)$ versus $(0,1)/(1,0)$ have equal marginals and disjoint support. This logical example is not a proper-coloring counterexample on an edge.

Proposition 10 (Symmetry and coupling)**.** *On a complete graph with exactly n colors, a vertex-equivariant stochastic protocol with exchangeable private inputs is uniform over proper labeled colorings conditional on successful termination of positive probability. Any coupling (X,Y) satisfies $d_{\mathrm{TV}}(\mathcal{L}(X),\mathcal{L}(Y)) \le \Pr(X \neq Y)$.*

Proof. Vertex permutations act transitively on the $n!$ proper coloring bijections; equivariance and the invariant success event make their conditional masses equal. For any event A, the difference in probabilities of $X, Y \in A$ is at most $\Pr(X \neq Y)$; take the supremum. □

Anonymity alone does not establish the symmetry premises. Untransformed IDs, non-equivariant schedules and numerical ordering require care. With more than n colors, vertex symmetry need not equate different color subsets. For r independent paired runs of one graph/policy protocol with zero disagreements, the one-sided upper limit $1-\alpha^{1/r}$ bounds disagreement and hence TV under that coupling. It says nothing universal about other graphs/checkpoints. Uniform conditional full-transition coupling error ϵ for H common pre-divergence steps gives $d_{\mathrm{TV}} \le H\epsilon$; per-node error would also require a node union bound. Native accuracy is not this uniform premise.

Global-law, local-palette, robust-statistics and factorial studies remain in their historical directories with schemas, fixed banks and testing families. Complete runs are observations, permutations act on rows rather than independently on vertices, and witness selection needs an independent split or valid joint permutation scheme (Gretton et al., 2012; Hemerik and Goeman, 2018; Phipson and Smyth, 2010; Holm, 1979). Reused banks are historical reanalysis, not independent new confirmation. Nonrejection does not establish equal laws or tolerance equivalence; power against localized alternatives matters. These studies support scoped distributional distinctions, not new MMD/permutation theory, and are outside the 810,000 denominator.

K Novelty Comparison and Reproduction Map

Candidate	Established precedent	Defensible increment
Neural coloring	Lemos et al. (2019); Langedal and Manne (2025); Vanderbush and Weber (2026); Le et al. (2026)	Seven-rule native coverage with declared information and complete selected-checkpoint provenance
Trace imitation; local rules	Veličković et al. (2020, 2022); Minder et al. (2023); Qharabagh et al. (2026)	Seven scoped teachers, learned/supplied accounting and contract-specific finite evaluation, not first learned distributed execution
Interactive/CF learning	Ross et al. (2011); de Haan et al. (2019); Geiger et al. (2022); Wu et al. (2022); Bevilacqua et al. (2023)	Specific admissibility contracts and paired continuation gaps, not a new general causal method or equal-label advantage
Output versus mechanism	Saldyt and Kambhampati (2025); Najafgholizadeh and Afsharchi (2026); Rodionov and Prokhorenkova (2025)	Component-level rescues, contract support and progress/repair mechanisms in coloring, not novelty of the distinction or verified general execution
Information/locality	Linial (1992); Loukas (2020); Sato et al. (2019)	JP representation control and interpreter accounting; the elementary observation lemma is not a new lower bound
Algorithm extrapolation	Xu et al. (2021); Mathys et al. (2023); Wittig et al. (2026)	Frozen selected-condition outcomes and replayable palette/preference failures, not a general extrapolation theorem
Randomized output laws	Gretton et al. (2012); Hemerik and Goeman (2018); Holm (1979)	Matched statistical objects and protocol-specific evidence, not new test theory

The contribution combines scoped empirical coverage with continuation contracts and replayable functional mechanisms; none of trace supervision, distributed neural execution, interventions or the distinction between success and fidelity is claimed individually new.

Reproducibility. The accompanying repository provides teacher interfaces, continuation oracles, training and evaluation drivers, checkpoint identities and evaluation namespaces. Each campaign has a separate protocol and artifact index. The drivers reproduce continuation queries, autonomous executions and component interventions with frozen models and fixed case generators. Code and experimental artifacts are available from the project repository and its linked releases; repository documentation gives the execution commands.

References

Leonid Barenboim and Michael Elkin. Distributed $(\Delta + 1)$-coloring in linear (in Δ) time, 2008. URL https://arxiv.org/pdf/0812.1379v2. Version 2; the implemented teacher uses only the degree-three base case.

Beatrice Bevilacqua, Kyriacos Nikiforou, Borja Ibarz, Ioana Bica, Michela Paganini, Charles Blundell, Jovana Mitrovic, and Petar Veličković. Neural algorithmic reasoning with causal regularisation. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2272–2288, 2023. URL https://proceedings.mlr.press/v202/bevilacqua23a.html.

C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

Pim de Haan, Dinesh Jayaraman, and Sergey Levine. Causal confusion in imitation learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/947018640bf36a2bb609d3557a285329-Abstract.html.

Assefaw Hadish Gebremedhin and Fredrik Manne. Scalable parallel graph coloring algorithms. *Concurrency: Practice and Experience*, 12:1131–1146, 2000. URL https://eecs.wsu.edu/~assefaw/publications/Concurrency12-2000.pdf.

Atticus Geiger, Zhengxuan Wu, Hanson Lu, Josh Rozner, Elisa Kreiss, Thomas Icard, Noah Goodman, and Christopher Potts. Inducing causal structure for interpretable neural networks. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 7324–7338, 2022. URL https://proceedings.mlr.press/v162/geiger22a.html.

Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012. URL https://www.jmlr.org/papers/v13/gretton12a.html.

Jesse Hemerik and Jelle Goeman. Exact testing with random permutations. *TEST*, 27:811–825, 2018. doi: 10.1007/s11749-017-0571-1.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6: 65–70, 1979. URL https://www.ime.usp.br/~abe/lista/pdf4R8xPVzCnX.pdf.

Öjvind Johansson. Simple distributed $\Delta+1$-coloring of graphs. *Information Processing Letters*, 70(5):229–232, 1999. doi: 10.1016/S0020-0190(99)00064-2. URL https://www.sciencedirect.com/science/article/abs/pii/S0020019099000642.

Mark T. Jones and Paul E. Plassmann. A parallel graph coloring heuristic. *SIAM Journal on Scientific Computing*, 14(3):654–669, 1993. doi: 10.1137/0914041. URL https://doi.org/10.1137/0914041.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. URL https://arxiv.org/abs/1412.6980.

Fabian Kuhn and Roger Wattenhofer. On the complexity of distributed graph coloring. In *Proceedings of the ACM Symposium on Principles of Distributed Computing*, 2006. URL https://ac.informatik.uni-freiburg.de/publications/publications/podc06.pdf.

Kenneth Langedal and Fredrik Manne. Graph neural networks as ordering heuristics for parallel graph coloring. In *2025 Proceedings of the Symposium on Algorithm Engineering and Experiments (ALENEX)*, pages 56–67. SIAM, 2025. doi: 10.1137/1.9781611978339.5.

Thien Le, Tianyu Zhao, and Melanie Weber. Contrastive neural algorithmic reasoning for graph coloring, 2026. URL https://arxiv.org/abs/2606.03923.

Henrique Lemos, Marcelo Prates, Pedro Avelar, and Luis Lamb. Graph colouring meets deep learning: Effective graph neural network models for combinatorial problems. In *2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI)*, pages 879–885, 2019. doi: 10.1109/ICTAI.2019.00125.

Nathan Linial. Locality in distributed graph algorithms. *SIAM Journal on Computing*, 21(1):193–201, 1992. URL https://courses.csail.mit.edu/6.852/08/papers/Linial.pdf.

Andreas Loukas. What graph neural networks cannot learn: Depth vs width. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/pdf?id=B1l2bp4YwS.

Michael Luby. Removing randomness in parallel computation without a processor penalty. *Journal of Computer and System Sciences*, 47(2):250–286, 1993. doi: 10.1016/0022-0000(93)90033-S. URL https://www.sciencedirect.com/science/article/pii/002200009390033S.

Joël Mathys, Florian Grötschla, Kalyan Varma Nadimpalli, and Roger Wattenhofer. Flood and echo net: Algorithmically aligned GNNs that generalize, 2023. URL https://arxiv.org/abs/2310.06970.

Julian Minder, Florian Grötschla, Joël Mathys, and Roger Wattenhofer. SALSA-CLRS: A sparse and scalable benchmark for algorithmic reasoning, 2023. URL https://arxiv.org/abs/2309.12253. Extended abstract presented at the Second Learning on Graphs Conference (LoG 2023).

Amin Najafgholizadeh and Mohsen Afsharchi. Fitting every hint is not enough: Premature floods in recurrent algorithmic reasoners, 2026. URL https://arxiv.org/abs/2610.14982. Concurrent work, 10 October 2026.

Belinda Phipson and Gordon K. Smyth. Permutation p-values should never be zero: calculating exact p-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1): Article 39, 2010. doi: 10.2202/1544-6115.1585.

Muhammad Fetrat Qharabagh, Artur Back de Luca, George Giapitzakis, and Kimon Fountoulakis. Learning to execute graph algorithms exactly with graph neural networks, 2026. URL https://arxiv.org/abs/2601.23207. Version 2, 5 June 2026.

Gleb Rodionov and Liudmila Prokhorenkova. Neural algorithmic reasoning without intermediate supervision. In *Advances in Neural Information Processing Systems*, volume 36, pages 51663–51674, 2023. URL https://papers.nips.cc/paper_files/paper/2023/file/a2370db7c99791ad5d9f3ef48ad6d464-Paper-Conference.pdf.

Gleb Rodionov and Liudmila Prokhorenkova. Discrete neural algorithmic reasoning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 51851–51865, 2025. URL https://proceedings.mlr.press/v267/rodionov25a.html.

Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, volume 15 of *Proceedings of Machine Learning Research*, pages 627–635, 2011. URL https://proceedings.mlr.press/v15/ross11a.html.

Lucas Saldyt and Subbarao Kambhampati. Mind the gap: Quantifying mechanistic gaps in algorithmic reasoning via neural compilation, 2025. URL https://arxiv.org/abs/2505.18623.

Ryoma Sato, Makoto Yamada, and Hisashi Kashima. Approximation ratios of graph neural networks for combinatorial problems. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://arxiv.org/abs/1905.10261.

Johannes Schneider and Roger Wattenhofer. A new technique for distributed symmetry breaking. In *Proceedings of the ACM Symposium on Principles of Distributed Computing*, 2010. URL https://tik-old.ee.ethz.ch/file/22a700b74fc2f37acb56d690e62c86cf/podcfp107_schneider_188.pdf.

Olga Solodova, Nick Richardson, Deniz Oktay, and Ryan P. Adams. Graph neural networks gone hogwild. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2407.00494.

Knut Vanderbush and Melanie Weber. Neural algorithmic reasoning for approximate k-coloring with recursive warm starts, 2026. URL https://arxiv.org/abs/2601.05137.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2017. URL https://arxiv.org/abs/1706.03762.

Petar Veličković and Charles Blundell. Neural algorithmic reasoning. *Patterns*, 2(7):100273, 2021. doi: 10.1016/j.patter.2021.100273.

Petar Veličković, Rex Ying, Matilde Padovano, Raia Hadsell, and Charles Blundell. Neural execution of graph algorithms. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/pdf?id=SkgK00EtvS.

Petar Veličković, Adrià Puigdomènech Badia, David Budden, Razvan Pascanu, Andrea Banino, Misha Dashevskiy, Raia Hadsell, and Charles Blundell. The CLRS algorithmic reasoning benchmark. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 22084–22102, 2022. URL https://proceedings.mlr.press/v162/velickovic22a.html.

Solveig Wittig, Antonis Vasileiou, Robert R. Nerem, Timo Stoll, Floris Geerts, Yusu Wang, and Christopher Morris. Which algorithms can graph neural networks learn?, 2026. URL https://arxiv.org/abs/2602.13106v2. Version 2, 23 August 2026.

Zhengxuan Wu, Atticus Geiger, Joshua Rozner, Elisa Kreiss, Hanson Lu, Thomas Icard, Christopher Potts, and Noah Goodman. Causal distillation for language models. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4288–4295, 2022. doi: 10.18653/v1/2022.naacl-main.318.

Keyulu Xu, Jingling Li, Mozhi Zhang, Simon S. Du, Ken-ichi Kawarabayashi, and Stefanie Jegelka. What can neural networks reason about? In *International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/1905.13211.

Keyulu Xu, Mozhi Zhang, Jingling Li, Simon S. Du, Ken-ichi Kawarabayashi, and Stefanie Jegelka. How neural networks extrapolate: From feedforward to graph neural networks. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=UH-cmocLJC.

Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/f22e4747da1aa27e363d86d40ff442fe-Abstract.html.